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(54) **CASSETTE INCLUDING PROTRUSION CONFIGURED TO CONTACT WIDTHWISE CENTER PORTION OF INK RIBBON TO CAUSE INK RIBBON TO BE CURVED**

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(52) **U.S. Cl.**

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(57) **ABSTRACT**

A cassette includes: a lower case having a box-like shape; a ribbon roll which is a roll of an ink ribbon; a take-up spool configured to take up the ink ribbon from the ribbon roll; and a protrusion provided in the lower case. The lower case includes: a front wall; a rear wall; a left wall; a right wall; a first inner wall; a second inner wall; a third inner wall; and a fourth inner wall. The protrusion protrudes rightward from a portion of the fourth inner wall other than a front edge and a rear edge of the fourth inner wall. The protrusion is configured to contact a widthwise center portion of the ink ribbon to cause the ink ribbon to be curved along a width of the ink ribbon so that the widthwise center portion of the ink ribbon protrudes rightward.

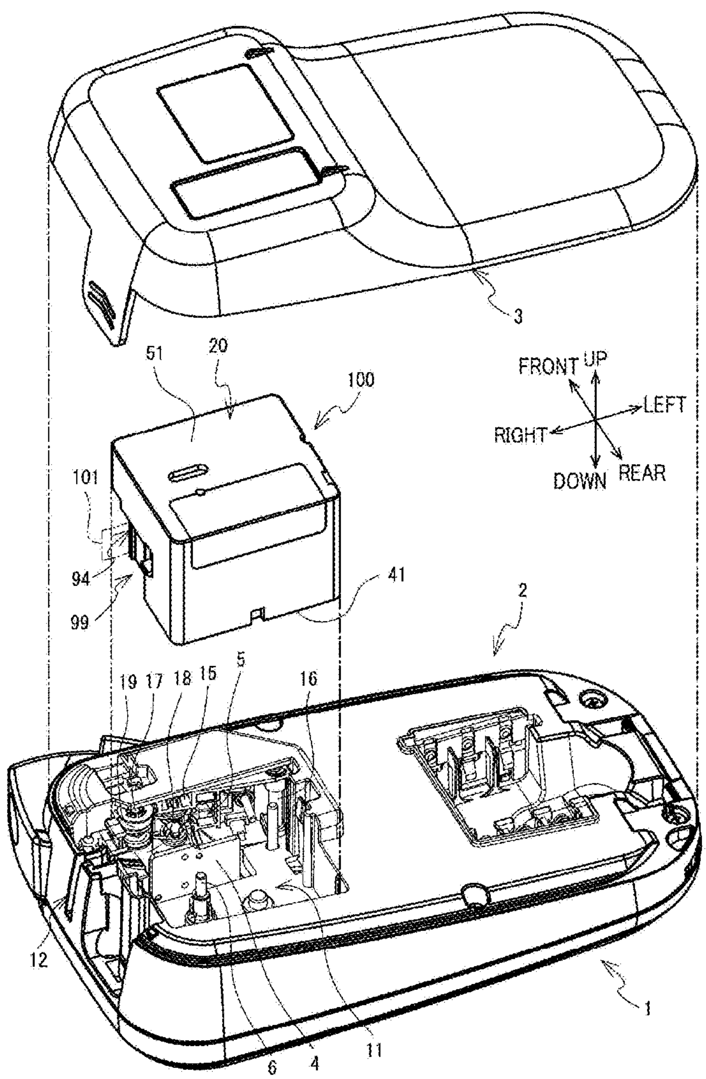
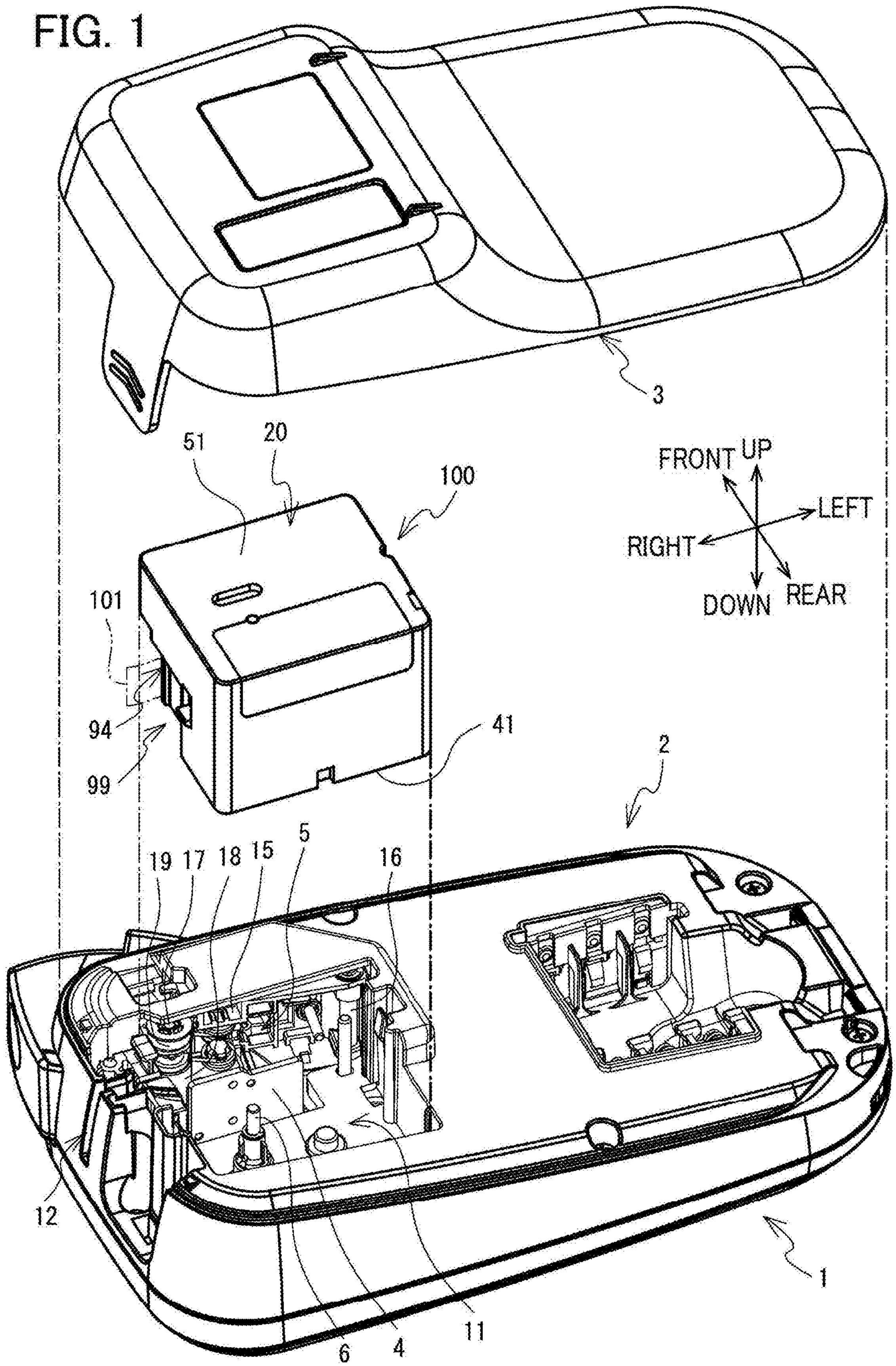


FIG. 1



3
51
20
100
FRONT
UP
LEFT
RIGHT
DOWN
REAR
101
94
99
41
2
19
17
18
15
5
16
12
1
6
4
11

FIG. 2

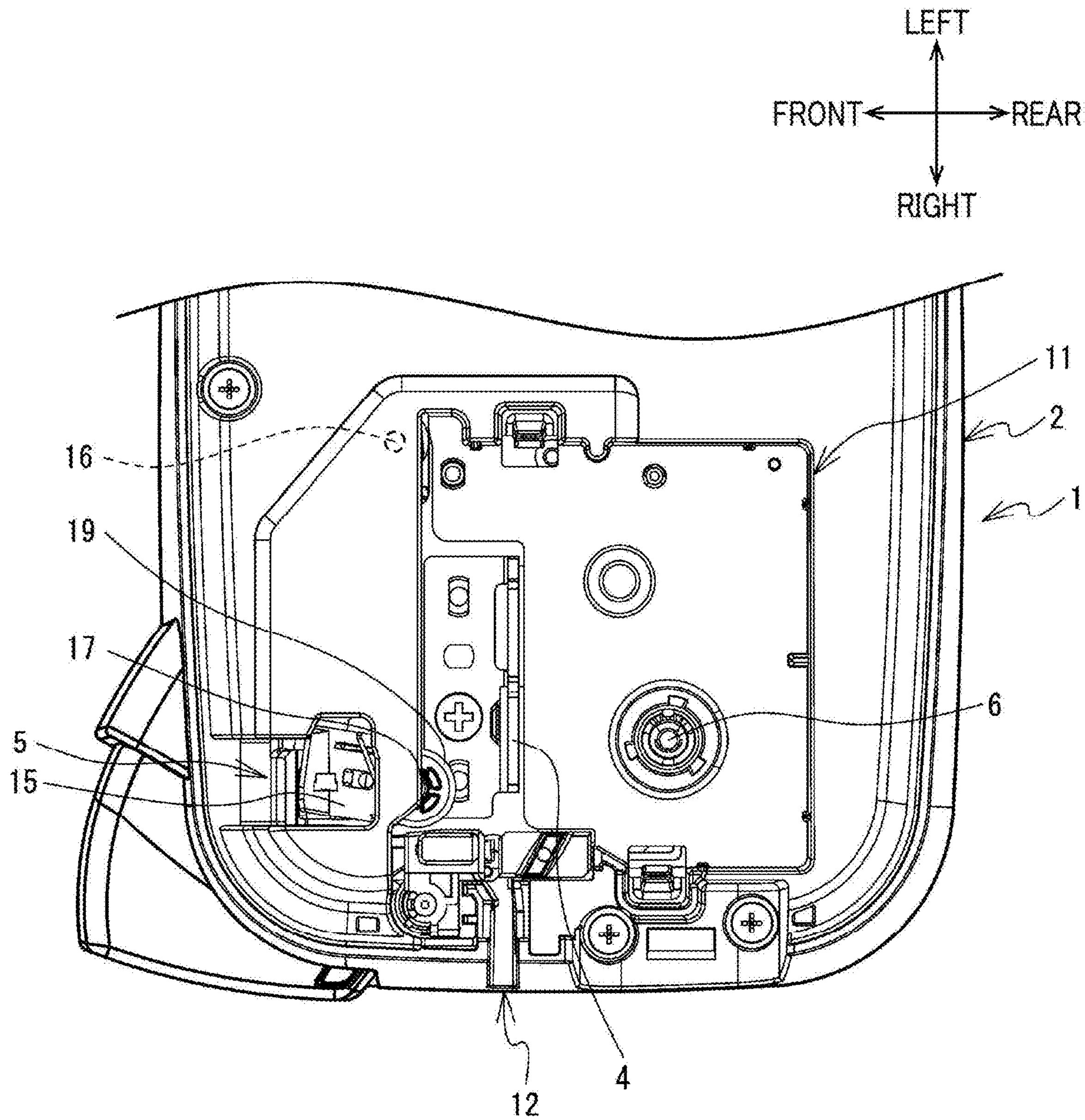


FIG. 3

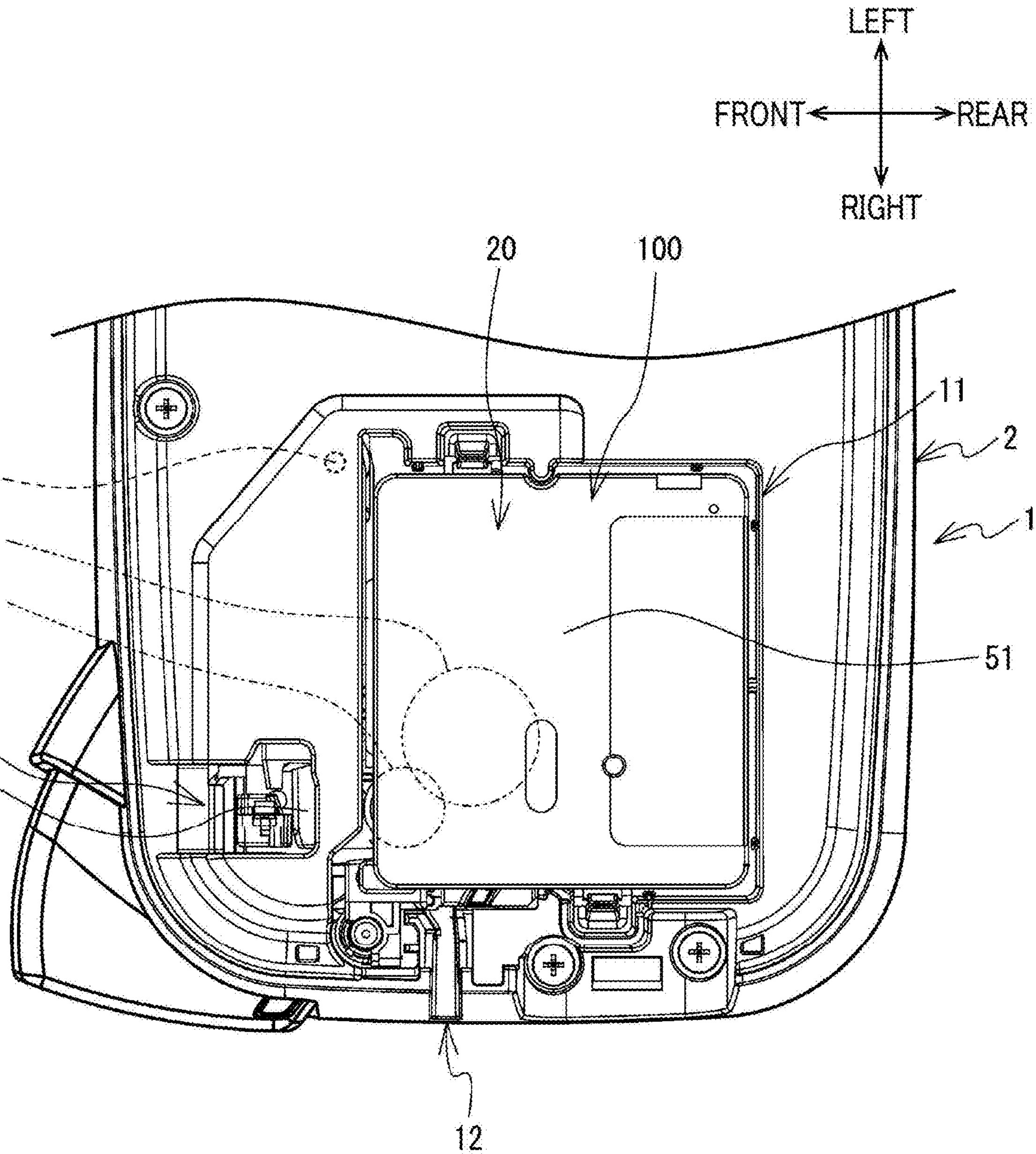


FIG. 4

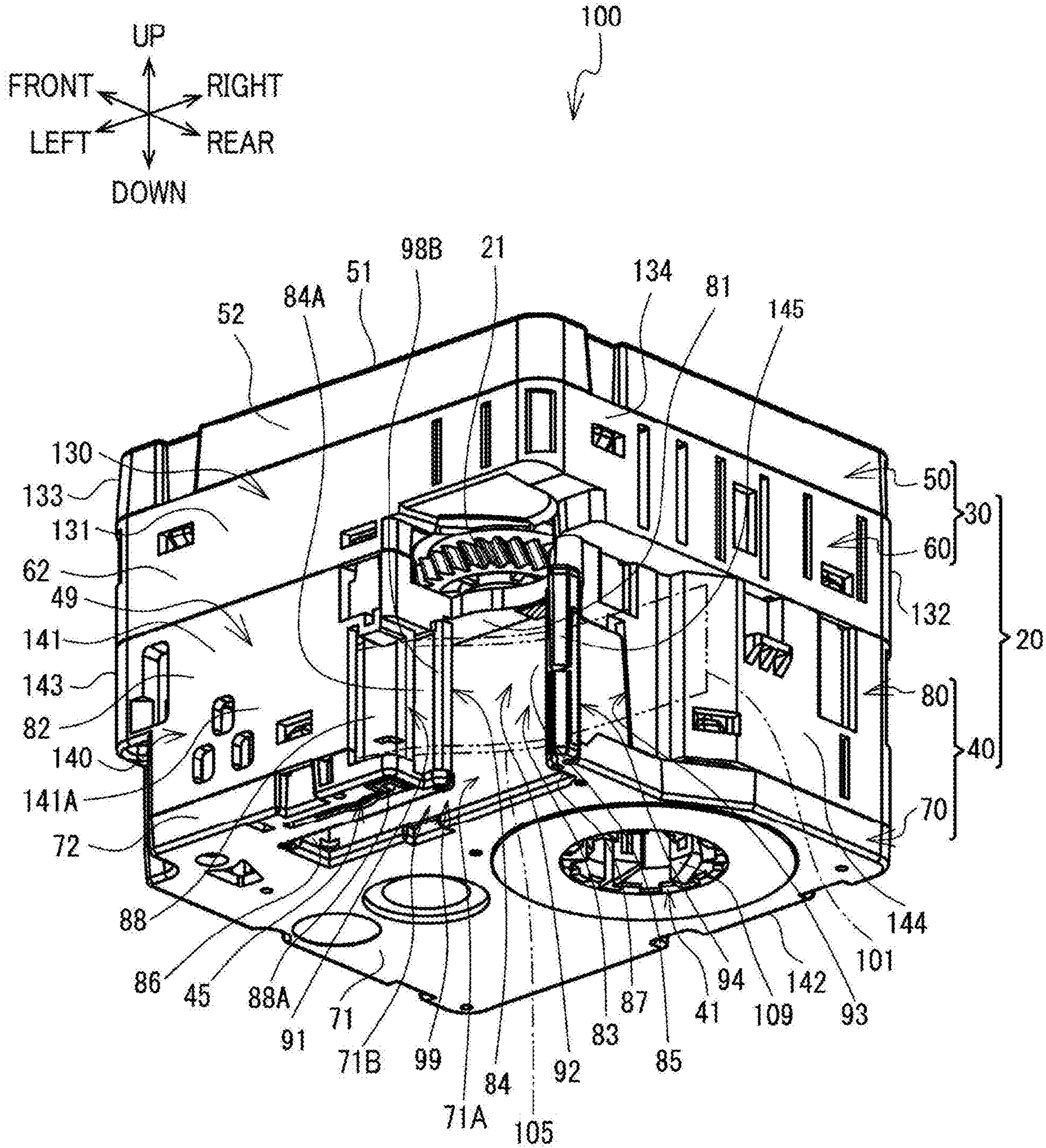
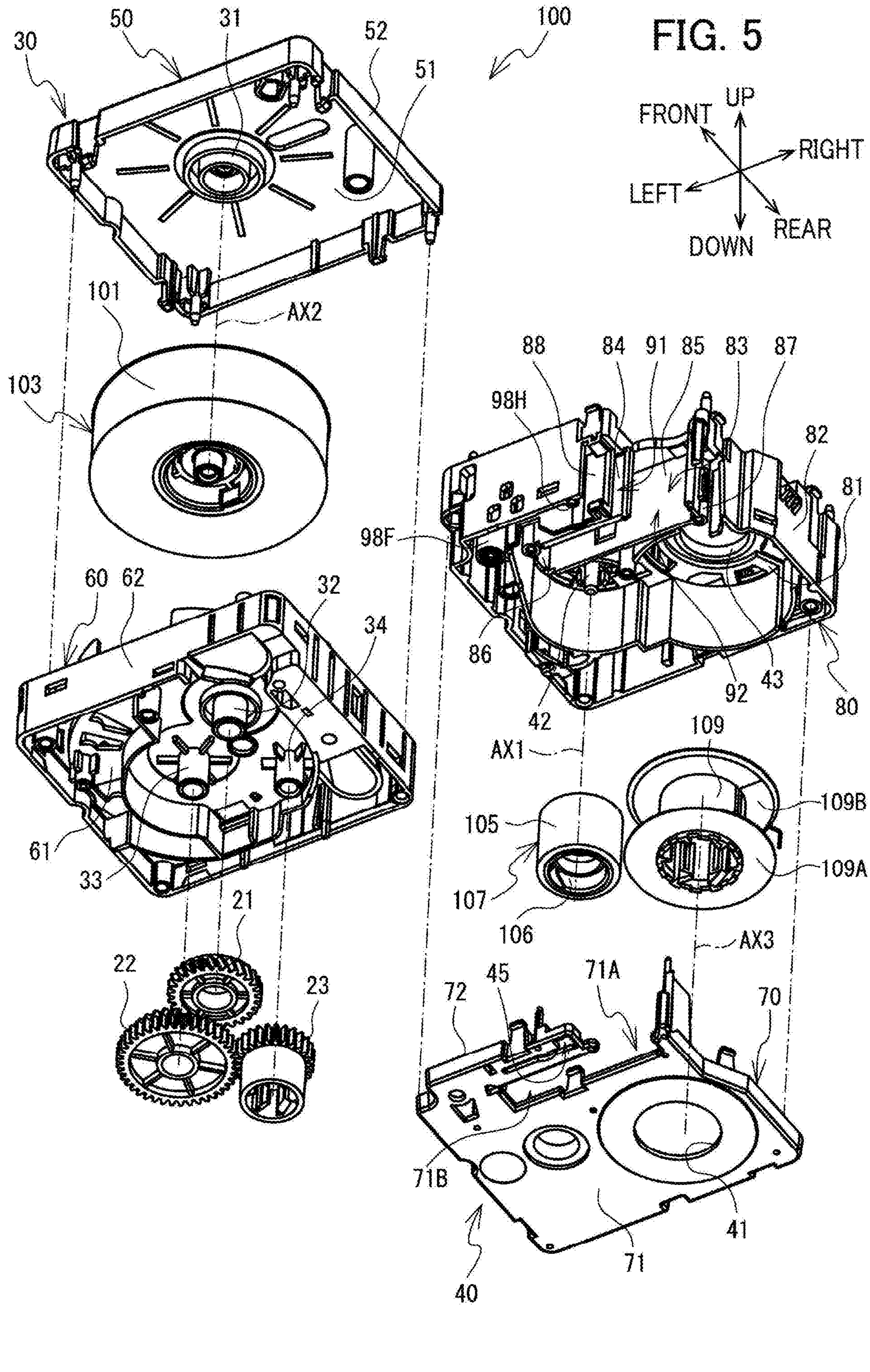

100
UP
FRONT
RIGHT
LEFT
REAR
DOWN
52
84A
51
98B
21
134
81
145
130
133
131
62
49
141
143
82
140
141A
72
88
86
45
88A
91
71
71B
99
71A
84
105
92
83
87
85
41
94
109
142
93
101
144
70
80
40
20
132
60
30
50

FIG. 5



30
50
31
52
51
100
UP
FRONT
RIGHT
LEFT
REAR
DOWN
101
AX2
103
88
84
91
85
83
87
98H
82
81
98F
60
62
32
34
86
92
43
80
42
AX1
109
109B
105
109A
61
33
107
106
21
22
23
AX3
45
71A
72
70
71B
41
71
40

FIG. 6

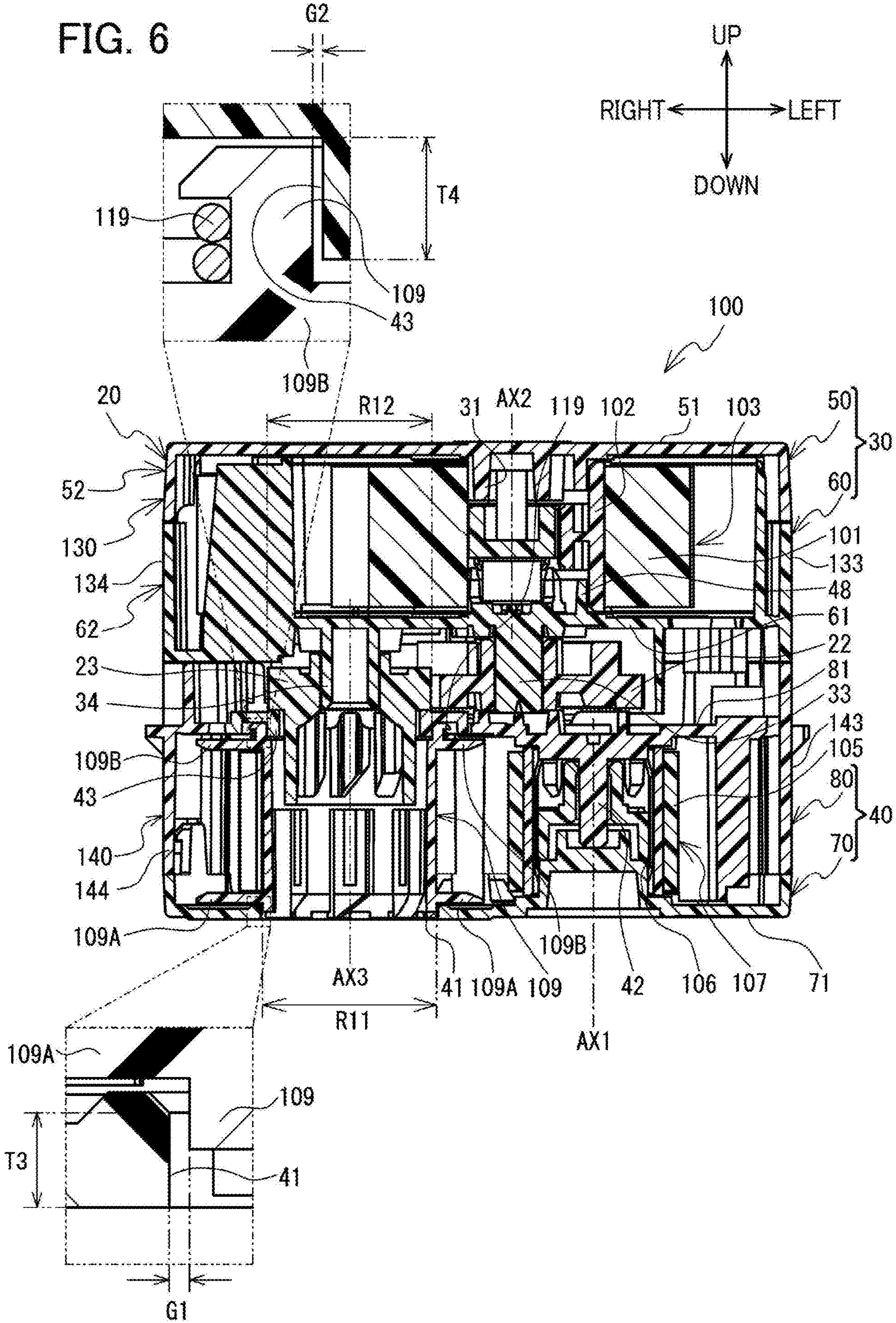

G2
UP
RIGHT
LEFT
DOWN
T4
119
109
43
109B
100
20
R12
31
AX2
119
102
51
103
50
30
60
52
130
134
62
23
34
109B
43
140
144
109A
101
133
48
61
22
81
33
143
105
80
40
70
AX3
41
109A
109
109B
42
106
107
71
R11
AX1
109A
109
T3
41
G1

FIG. 7

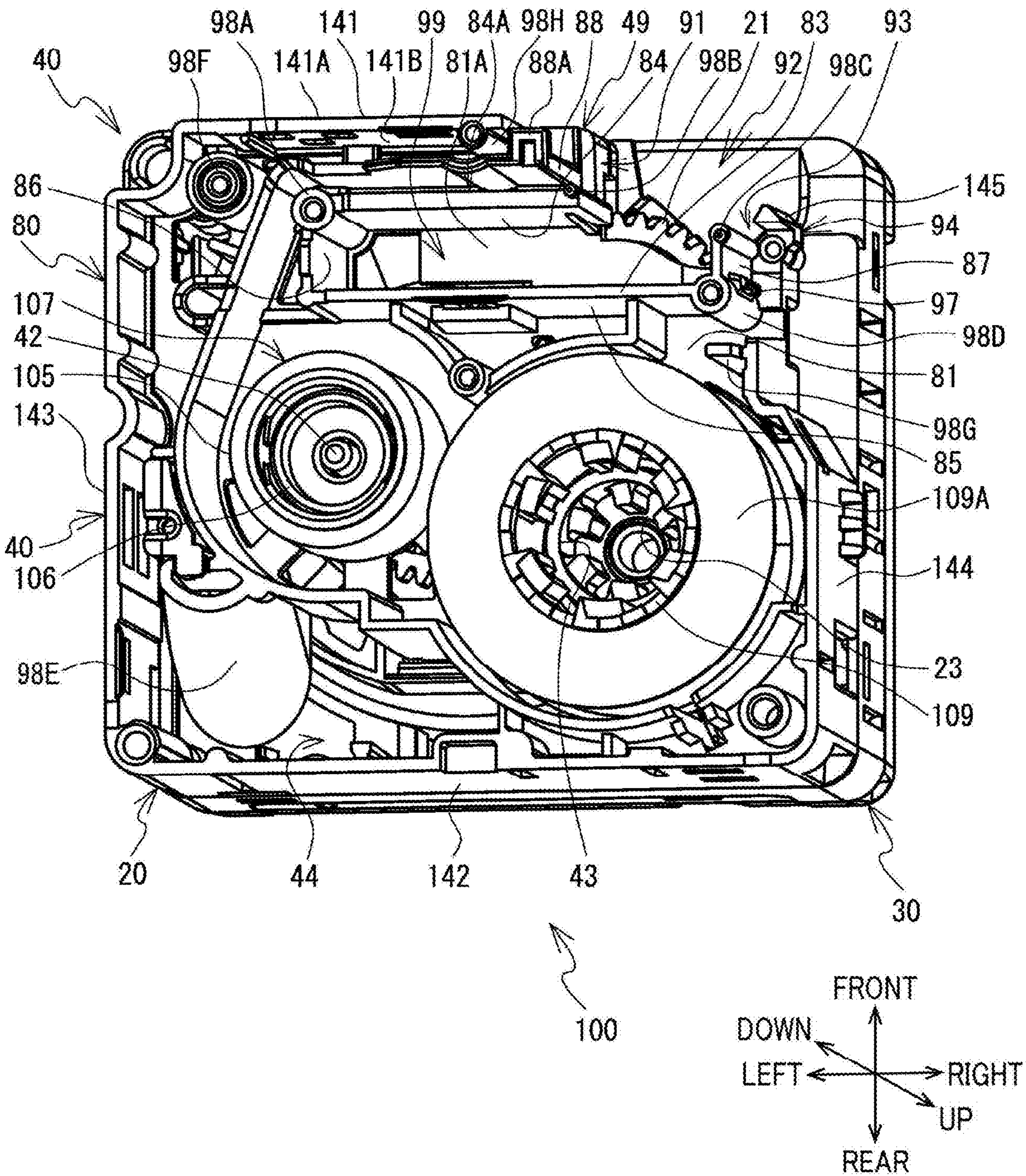

98A
141
99
84A
98H
88
49
91
21
83
93
40
98F
141A
141B
81A
88A
84
98B
92
98C
86
145
94
80
87
107
97
42
98D
105
81
143
98G
85
109A
40
106
144
98E
23
109
20
44
142
43
30
100
FRONT
DOWN
LEFT
RIGHT
UP
REAR

FIG. 8

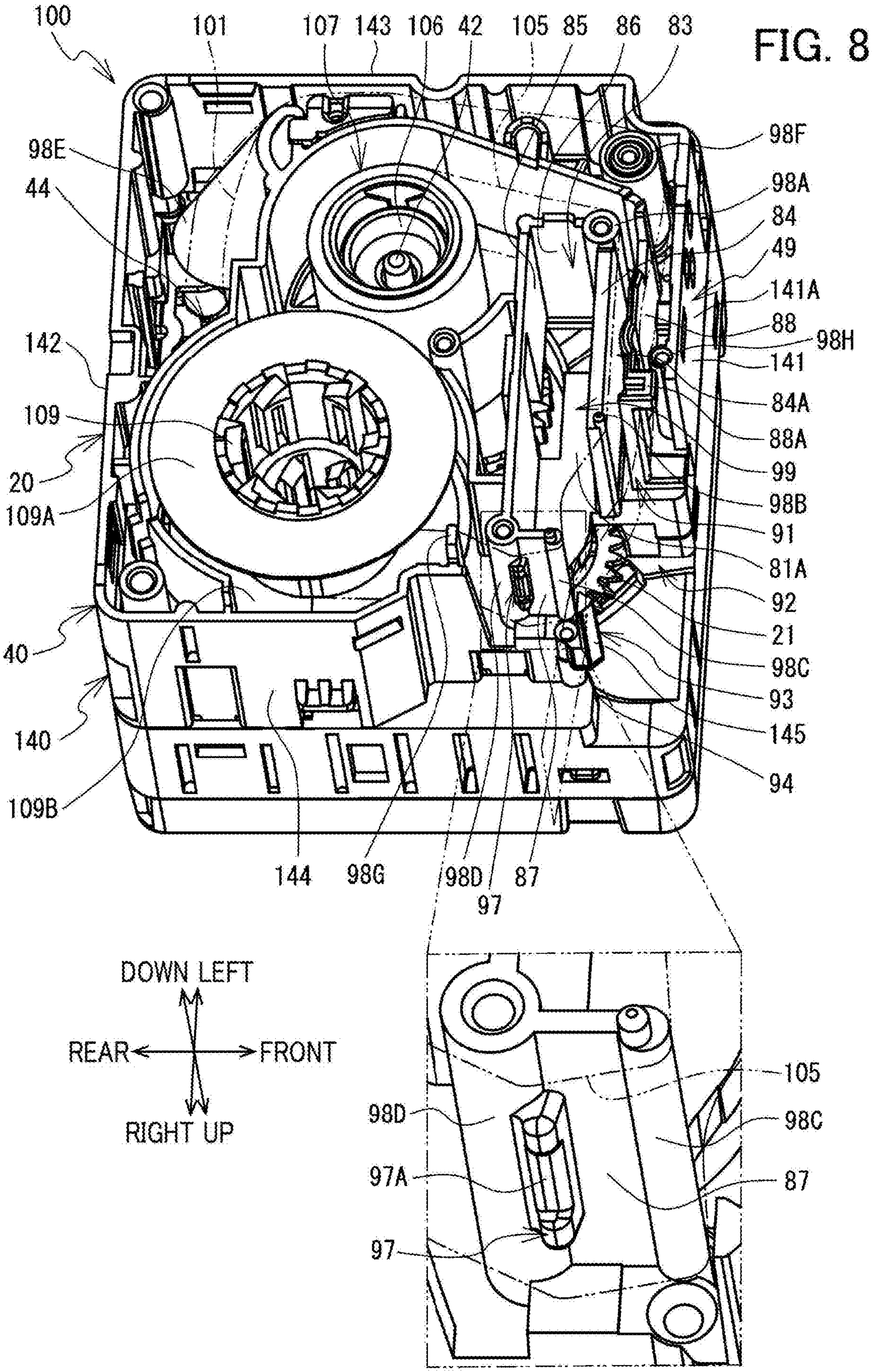

100
101
107
143
106
42
105
85
86
83
98F
98E
44
98A
84
49
141A
88
98H
142
141
84A
109
88A
20
99
109A
98B
91
81A
92
40
21
98C
93
140
145
94
109B
144
98G
98D
97
87
DOWN LEFT
REAR
FRONT
RIGHT UP
105
98D
98C
97A
87
97

FIG. 9

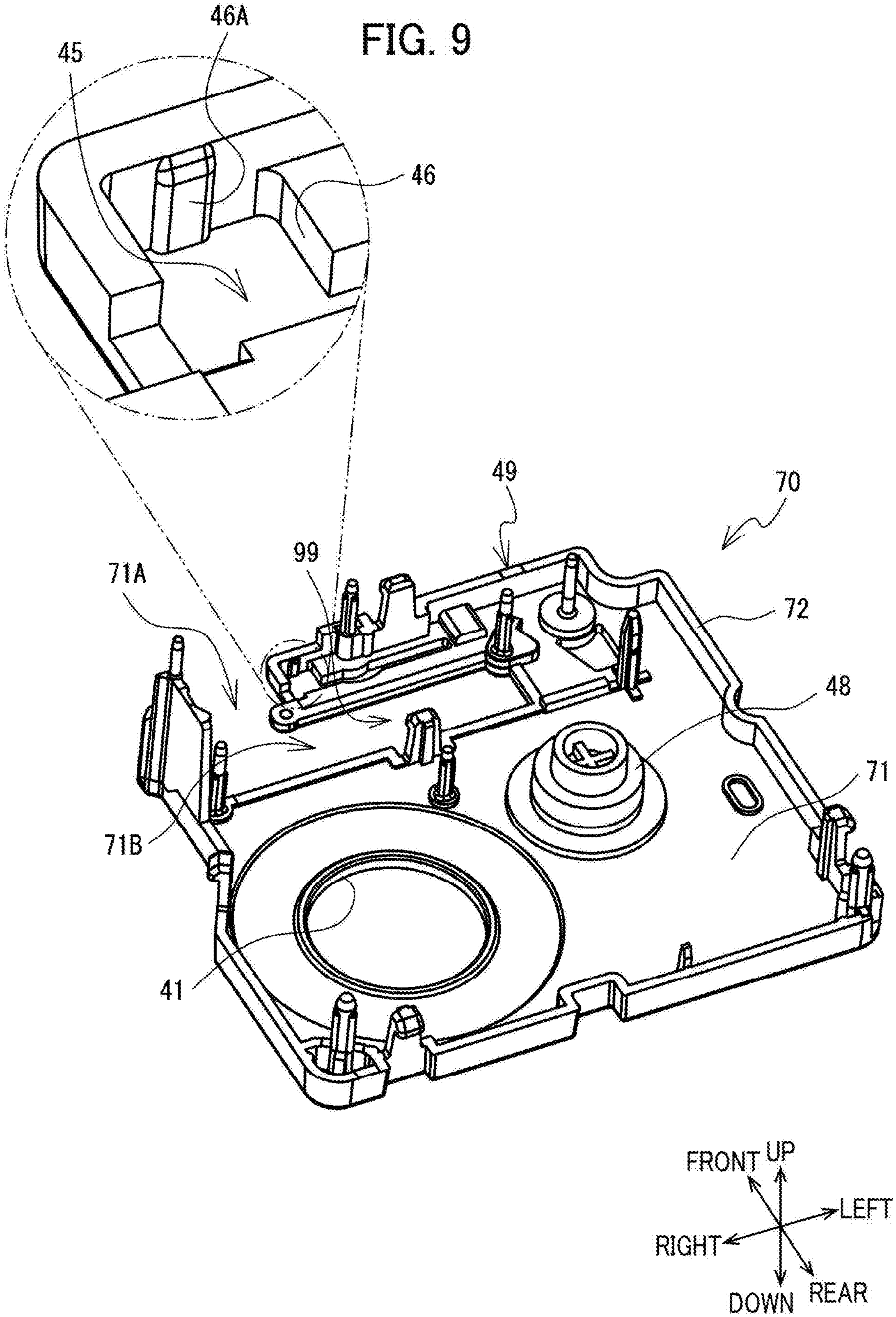

46A
45
46
49
70
99
71A
72
48
71
71B
41
FRONT
UP
LEFT
RIGHT
DOWN
REAR

FIG. 10

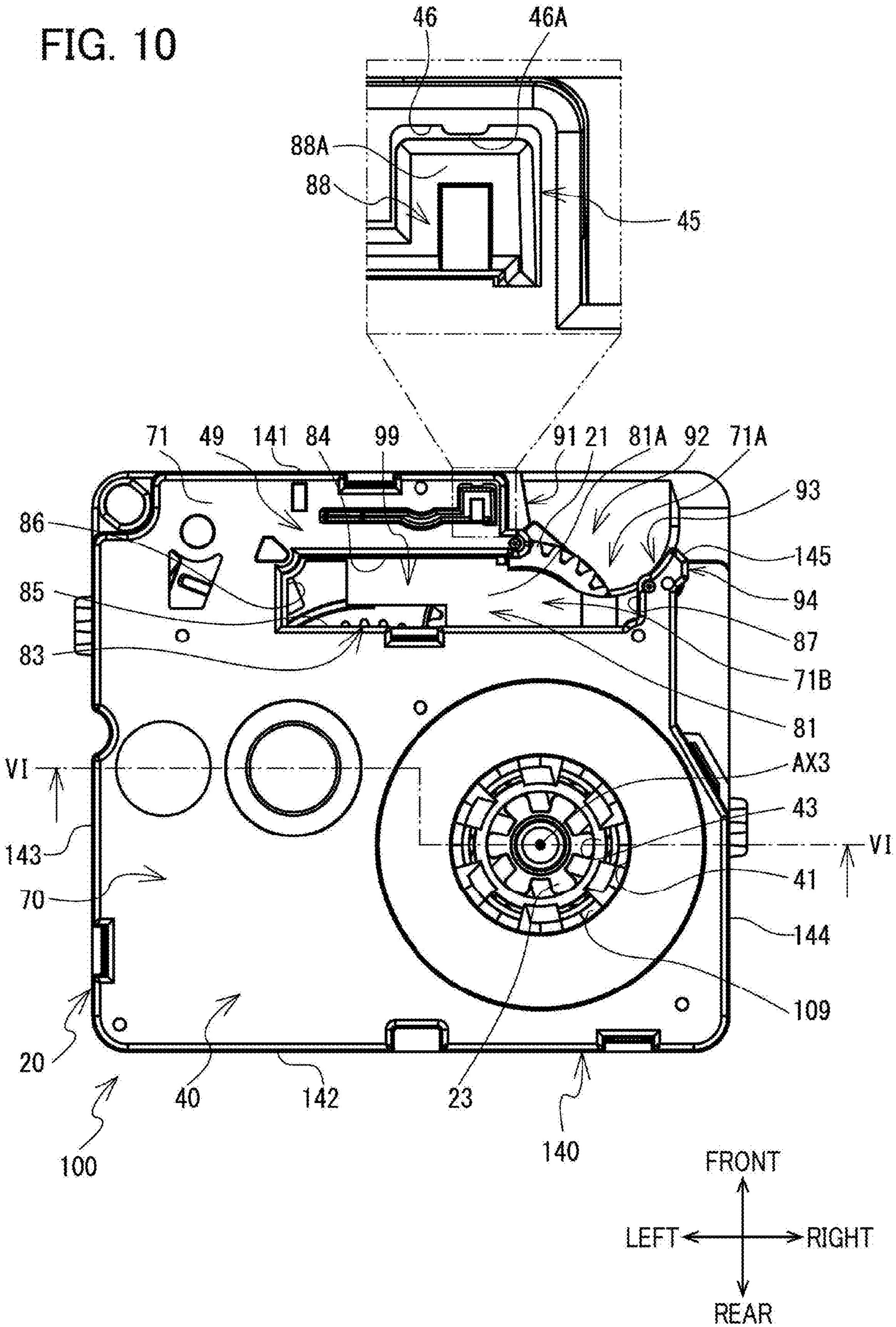

46
46A
88A
88
45
71
49
141
84
99
91
21
81A
92
71A
93
86
145
85
94
87
83
71B
81
VI
AX3
43
143
VI
41
70
144
109
20
40
142
23
140
100
FRONT
LEFT
RIGHT
REAR

FIG. 11

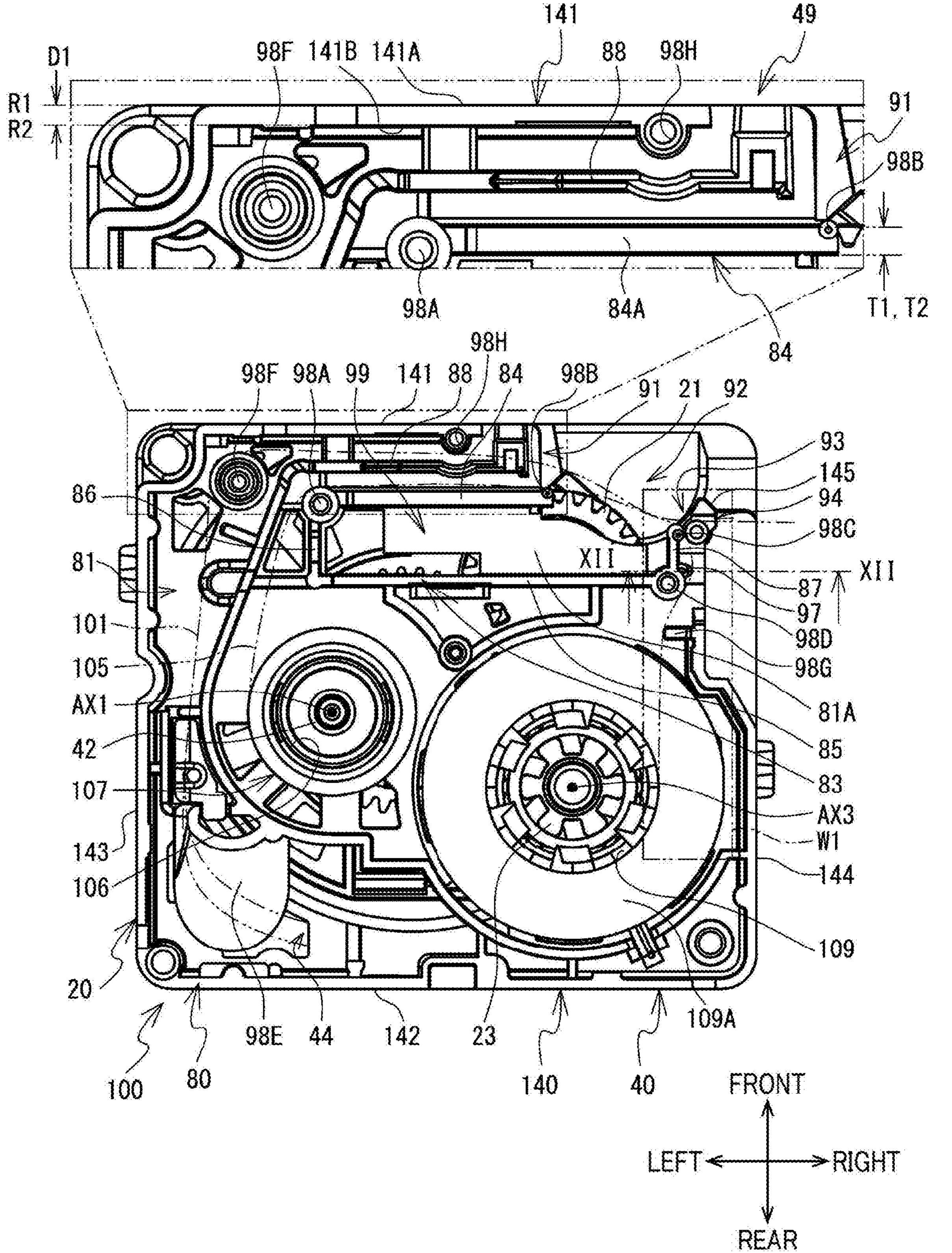

141
49
D1
98F 141B 141A
88
98H
R1
R2
91
98B
98A
84A
T1, T2
84
98H
98F 98A 99 141 88 84 98B
91 21 92
93
145
94
86
98C
81
XII
XII
87
97
98D
101
98G
105
AX1
81A
42
85
83
107
AX3
W1
143
144
106
109
20
109A
98E 44 142 23
100
80
140
40
FRONT
LEFT
RIGHT
REAR

FIG. 12

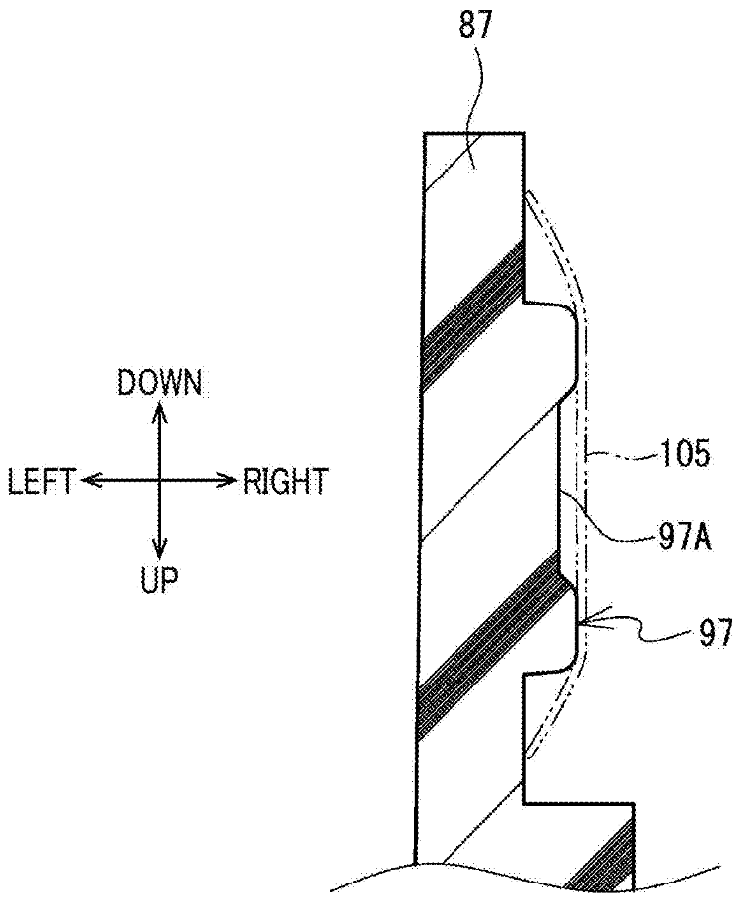


FIG. 13

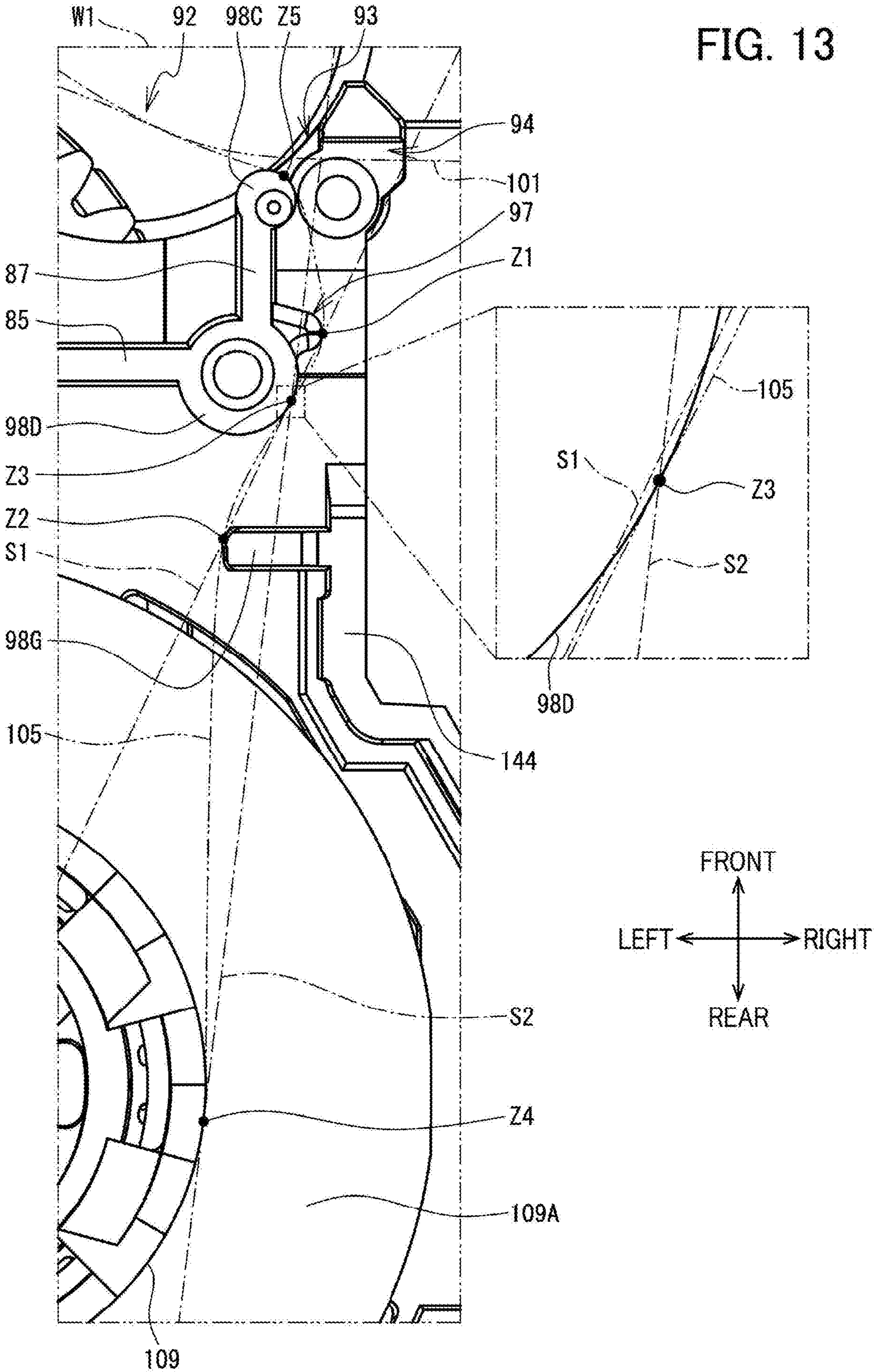

W1
92
98C
Z5
93
94
101
97
Z1
87
85
98D
Z3
Z2
S1
98G
105
144
S2
Z4
109A
109
105
S1
Z3
S2
98D
FRONT
LEFT
RIGHT
REAR

FIG. 14A

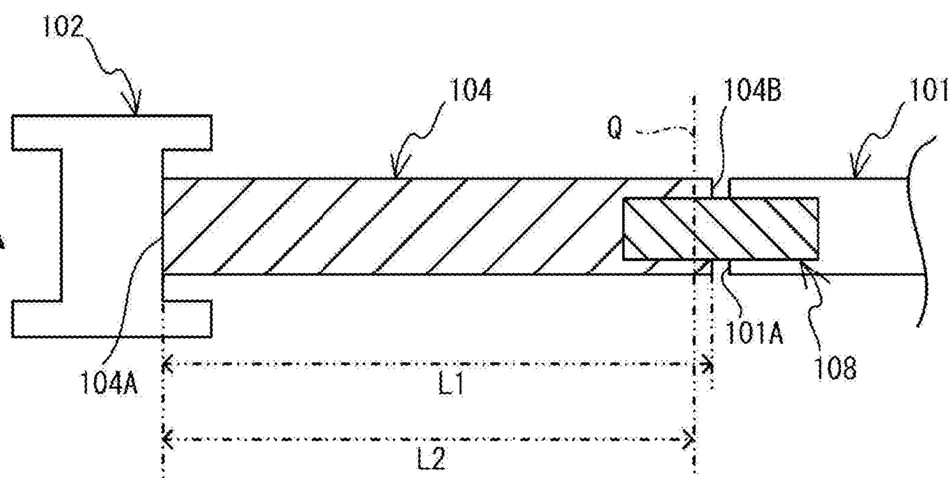


FIG. 14B

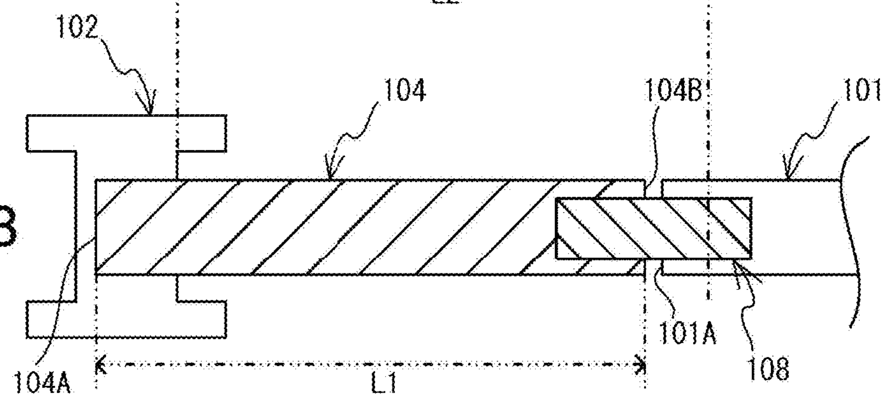


FIG. 15A

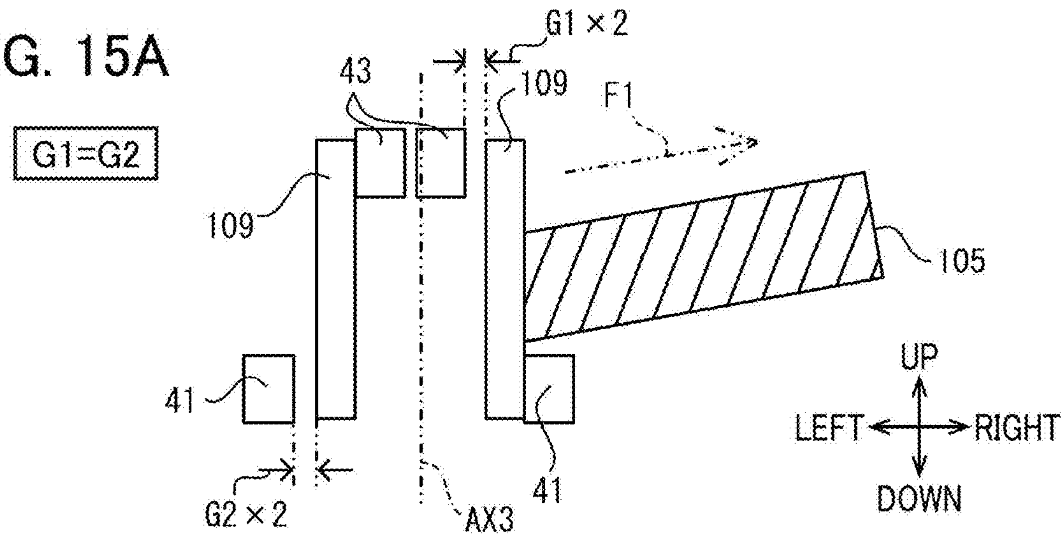


FIG. 15B

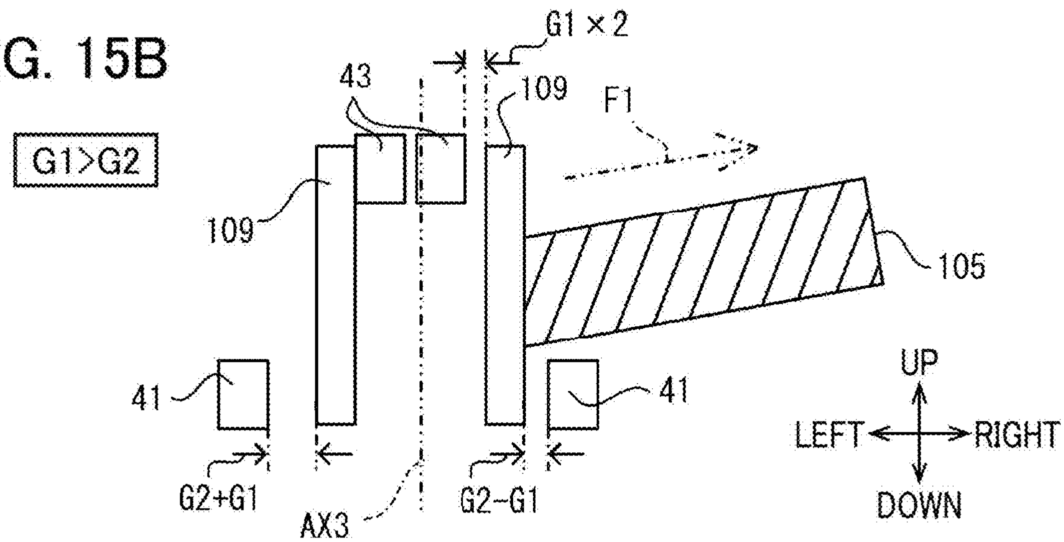


FIG. 15C

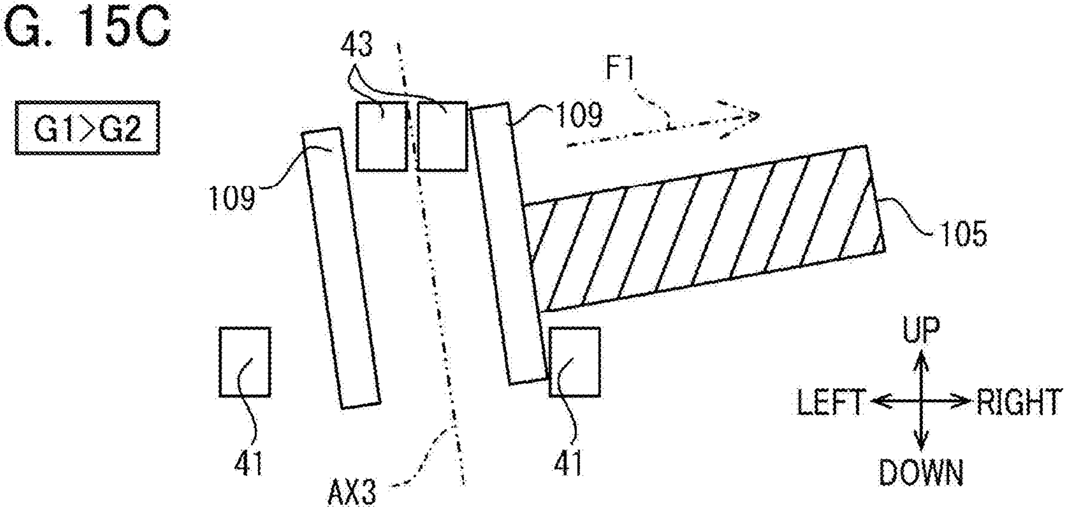


FIG. 16

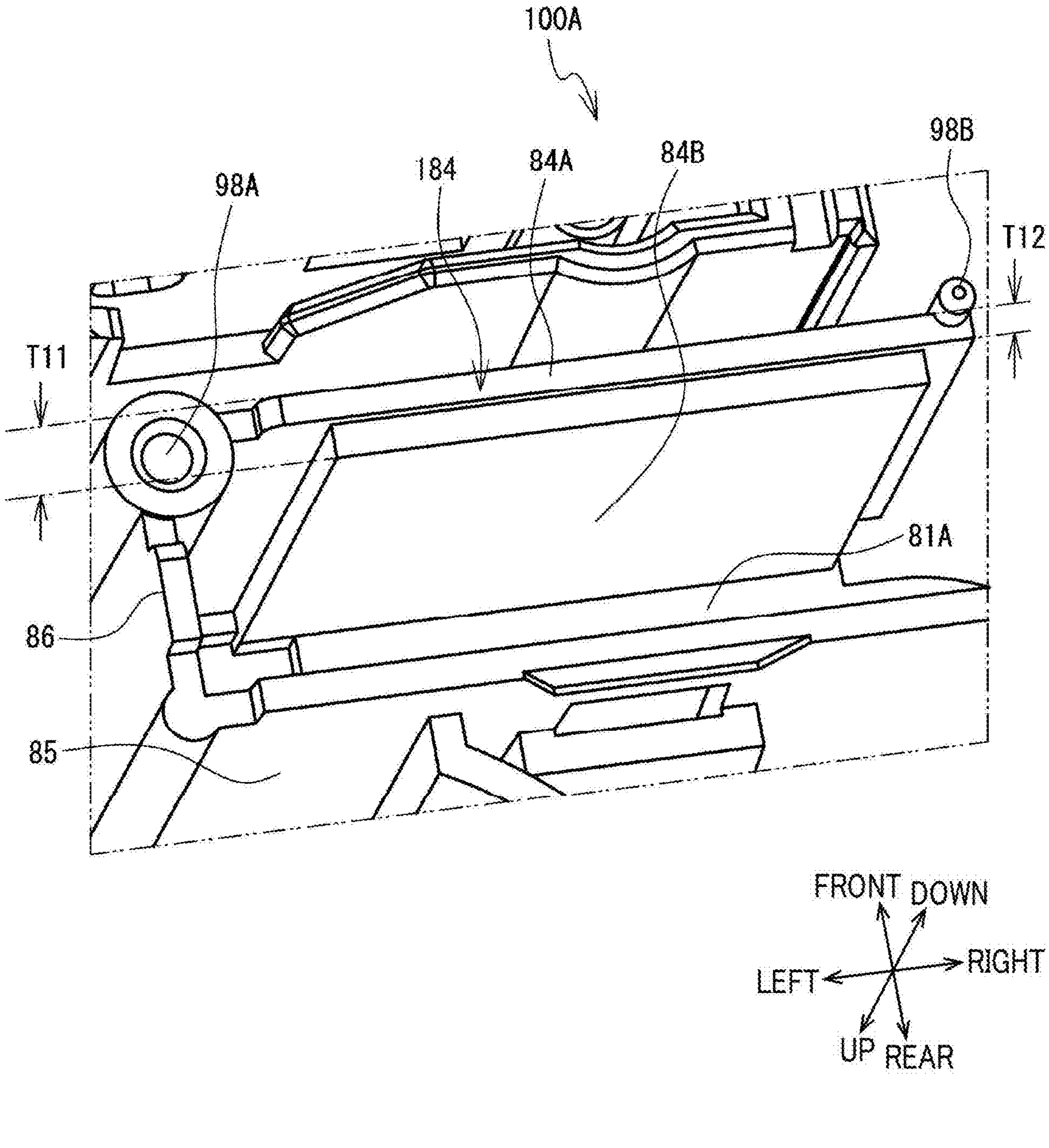

100A
98B
T12
184
84A
84B
98A
T11
81A
86
85
FRONT
DOWN
LEFT
RIGHT
UP
REAR

FIG. 17

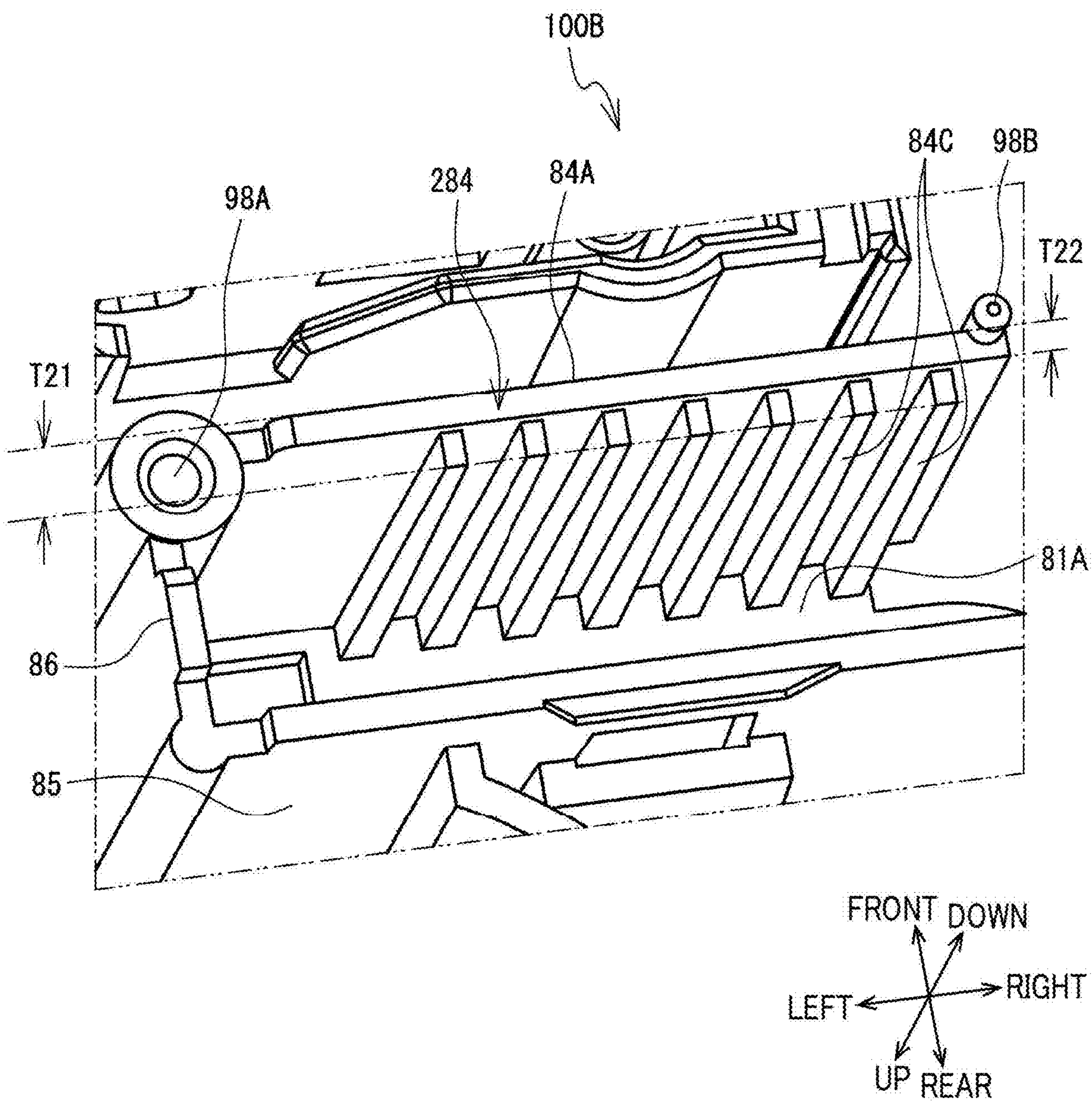
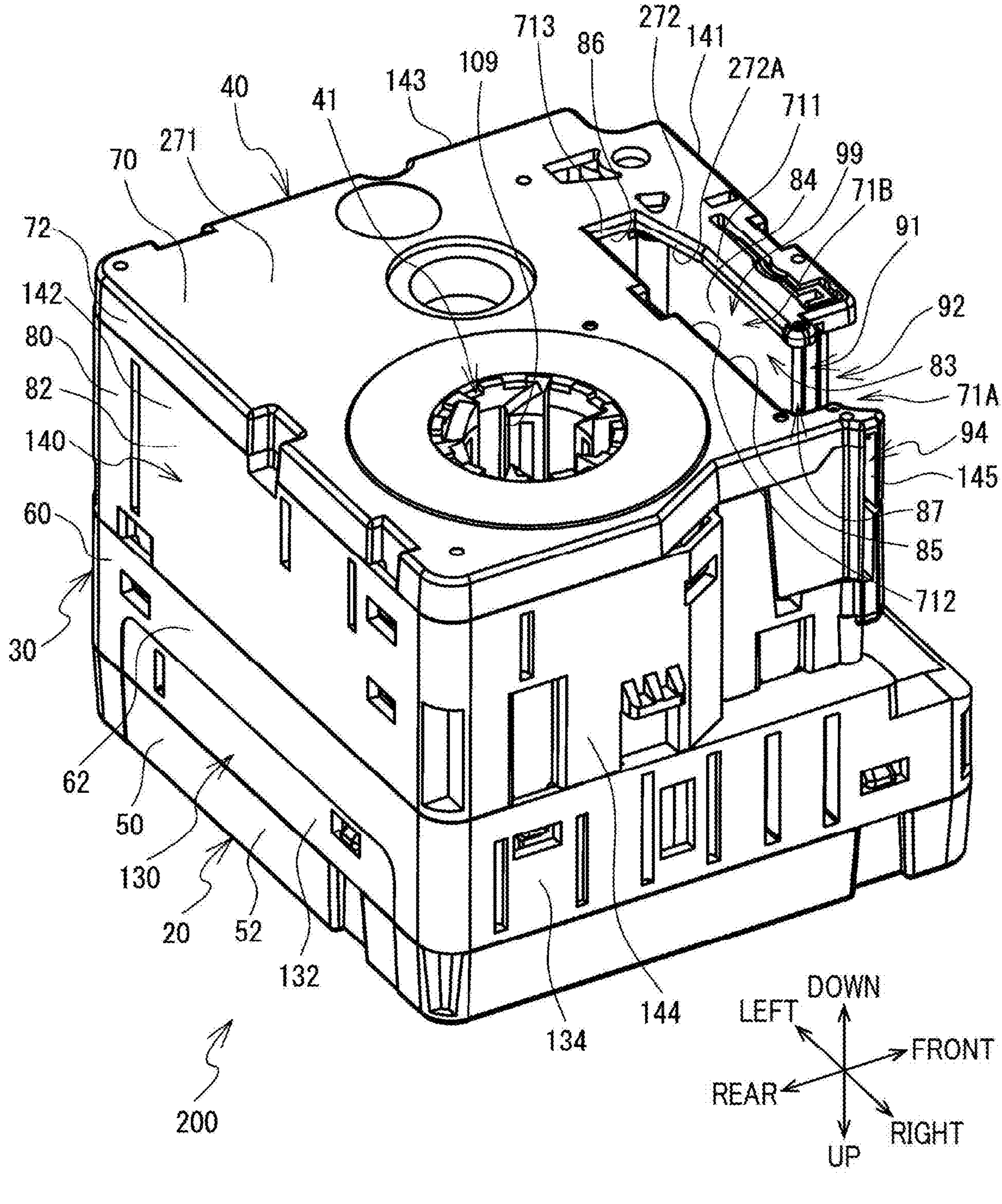

100B
84C
98B
284
84A
98A
T22
T21
81A
86
85
FRONT
DOWN
LEFT
RIGHT
UP
REAR

FIG. 18



40
70
271
41
143
109
713
86
272
141
272A
711
99
84
71B
91
92
83
71A
94
145
87
85
712
72
142
80
82
140
60
30
62
50
130
20
52
132
134
144
200
DOWN
LEFT
FRONT
REAR
RIGHT
UP

FIG. 19

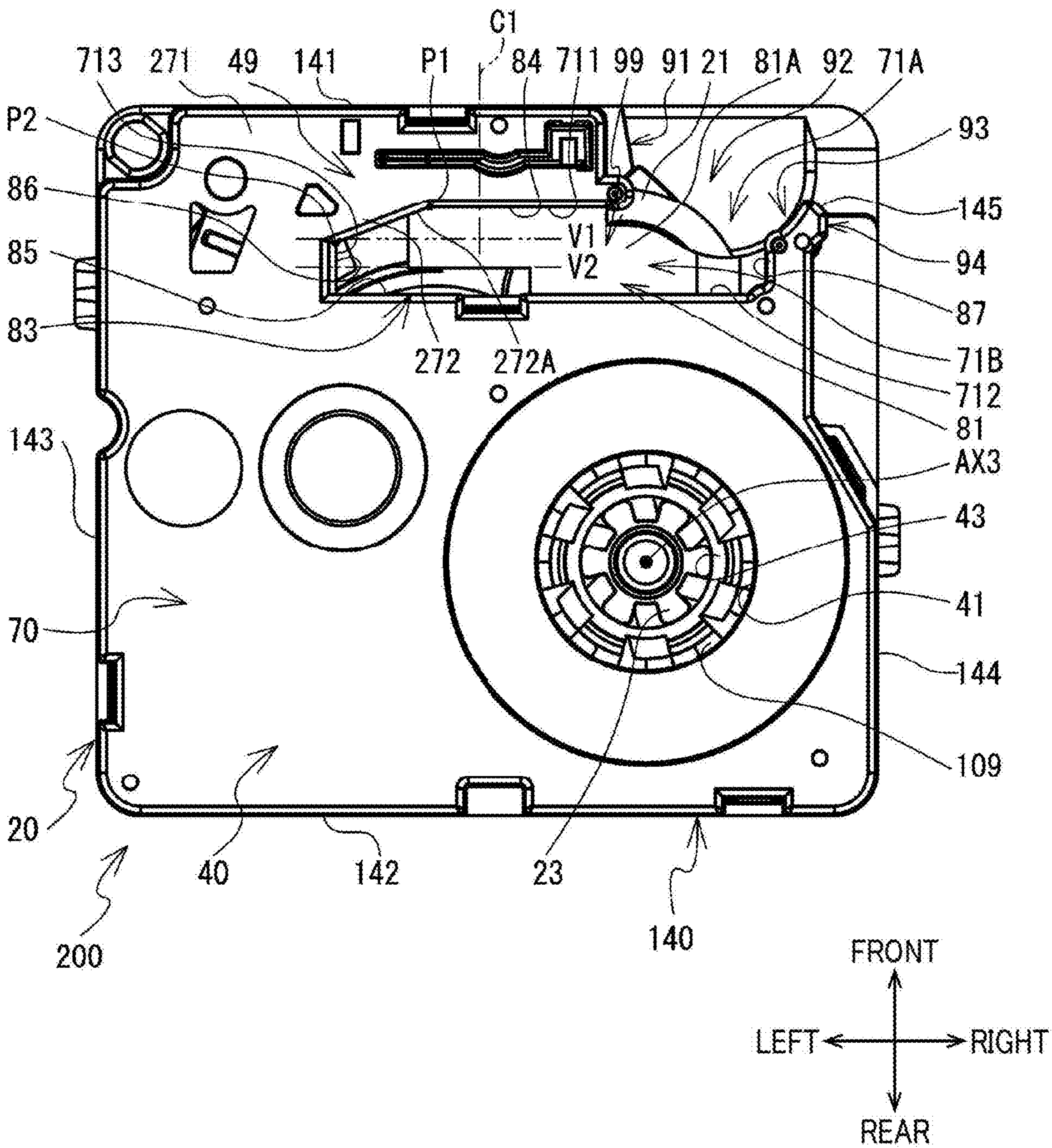

C1
713
271
49
141
P1
84
711
99
91
21
81A
92
71A
P2
93
86
145
V1
V2
85
94
87
83
272
272A
71B
712
81
143
AX3
43
41
70
144
109
20
40
142
23
200
140
FRONT
LEFT
RIGHT
REAR

FIG. 20

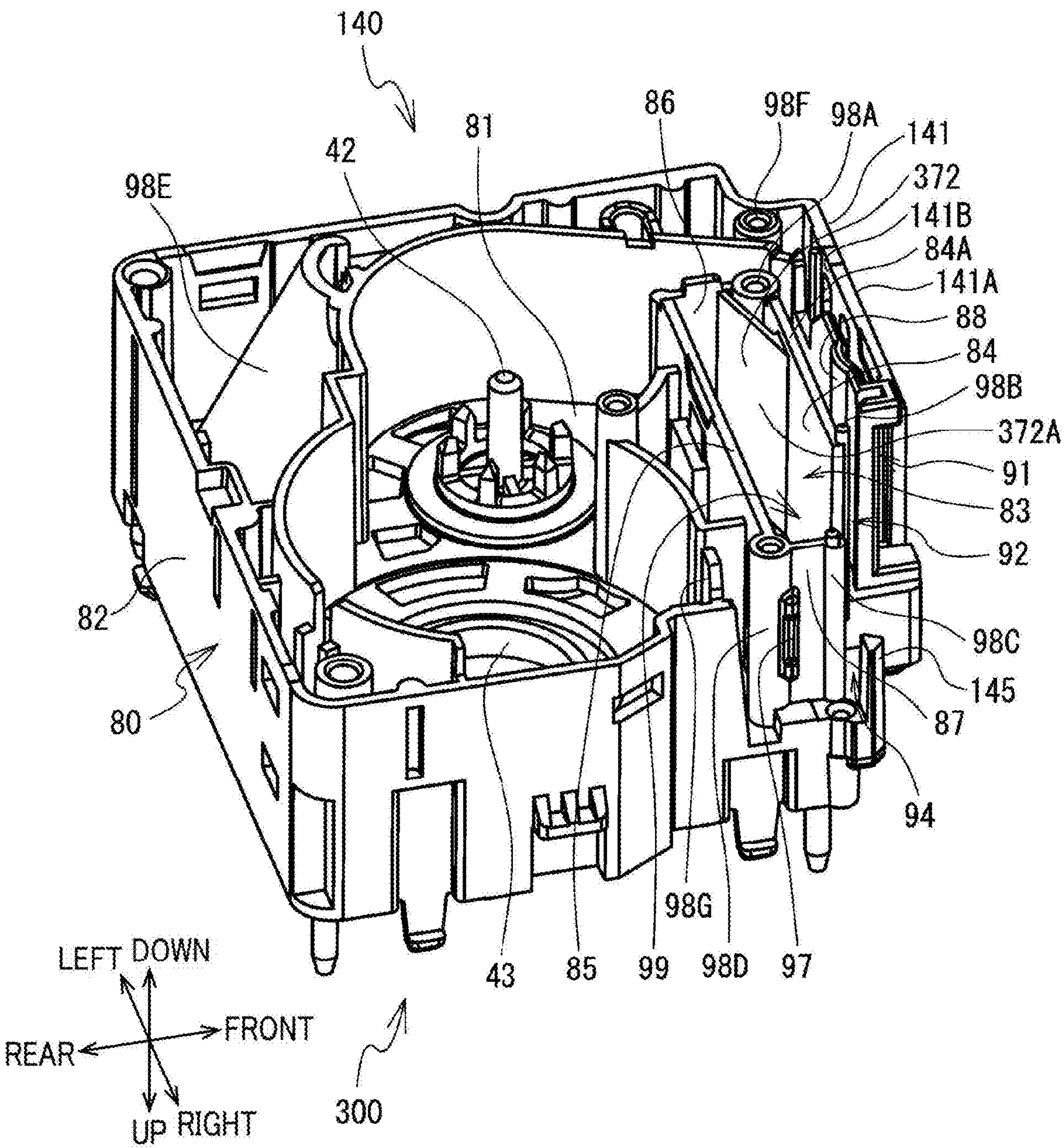

140
86
98F
98A
141
81
42
372
141B
98E
84A
141A
88
84
98B
372A
91
83
92
82
98C
145
80
87
94
98G
43
85
99
98D
97
LEFT
DOWN
FRONT
REAR
UP
RIGHT
300

FIG. 21

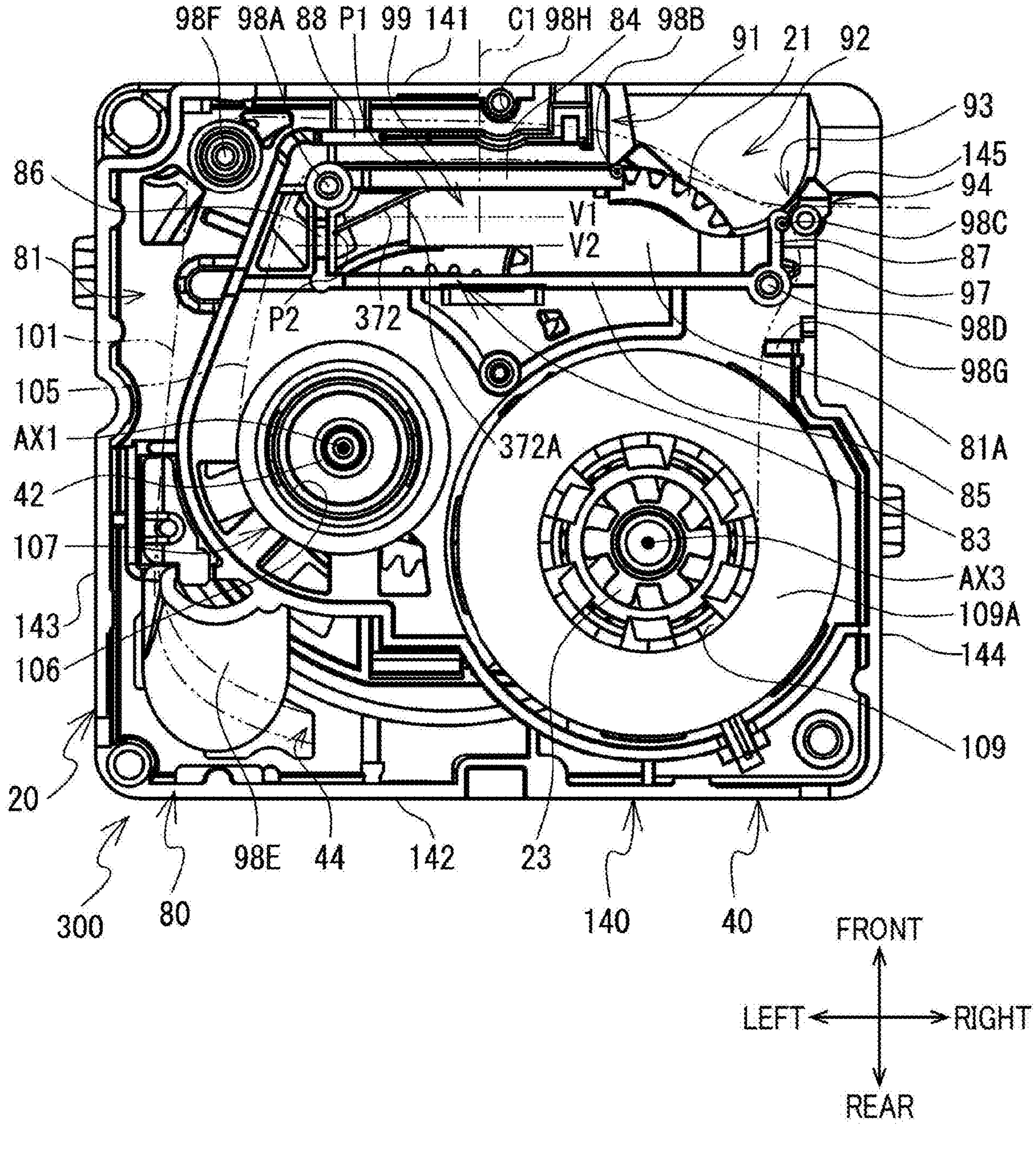
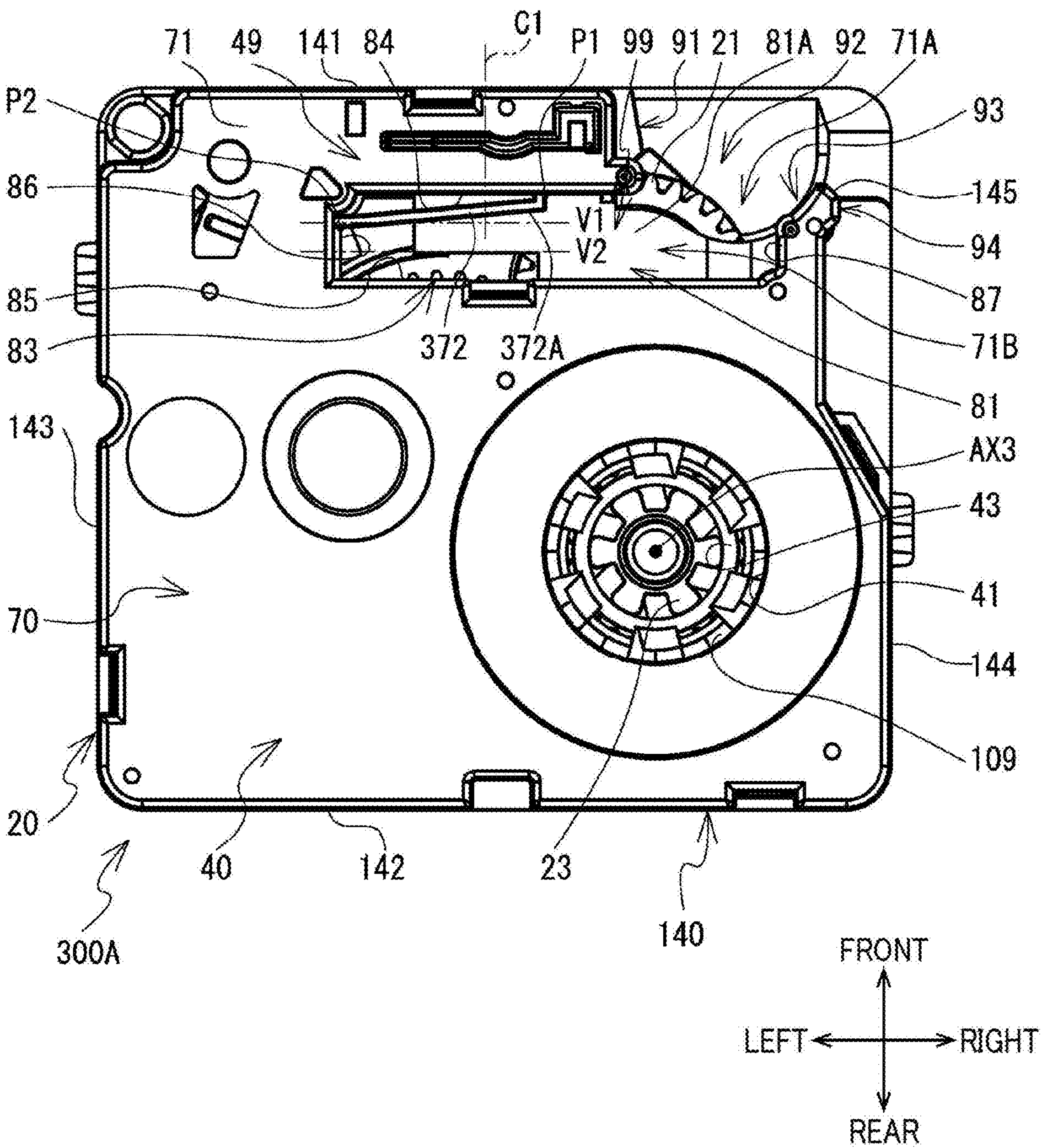
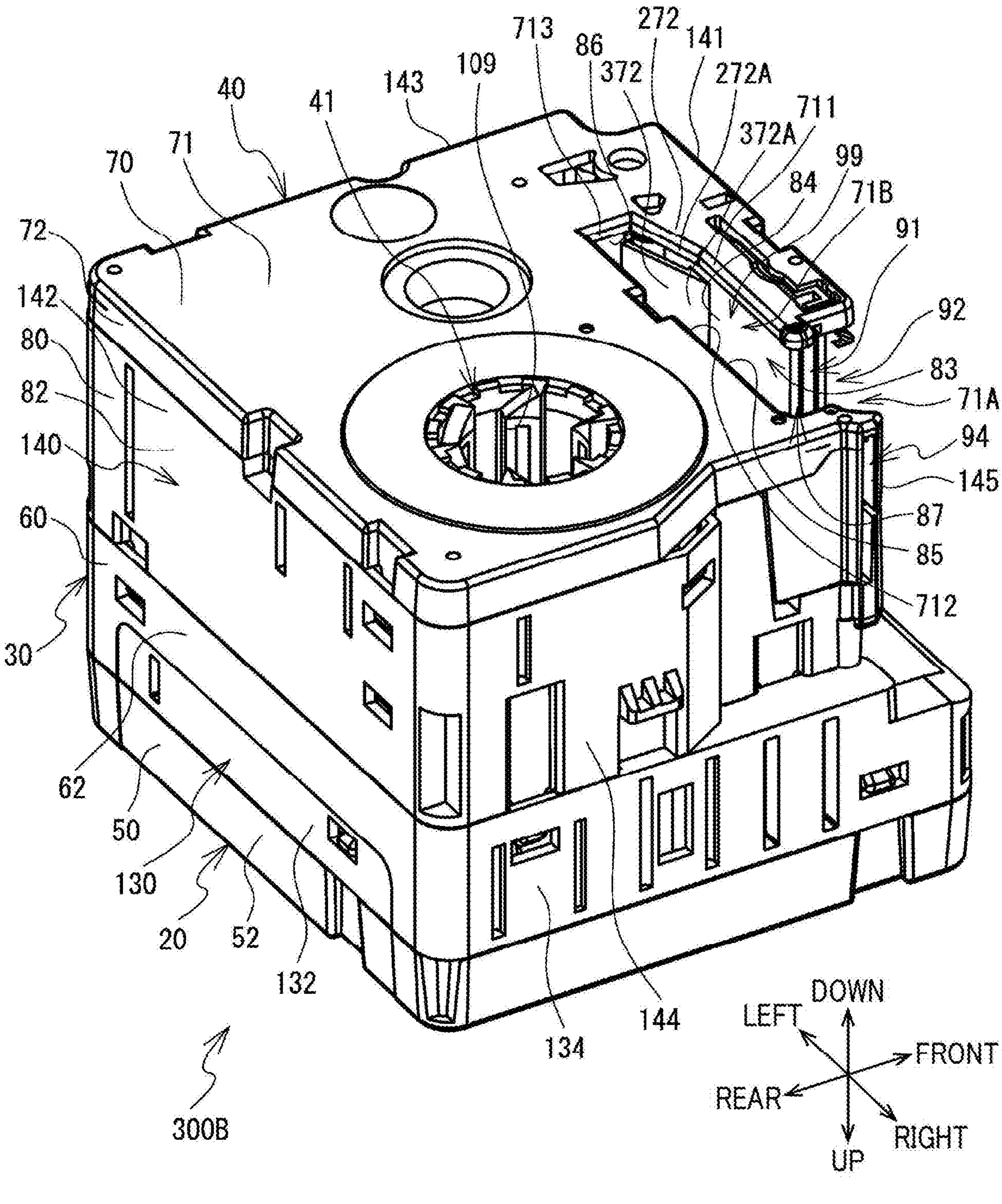

98F
98A
88
P1
99
141
C1
98H
84
98B
91
21
92
93
145
94
98C
87
97
98D
98G
81A
85
83
AX3
109A
144
109
86
81
101
105
AX1
42
107
143
106
20
300
80
98E
44
142
23
140
40
V1
V2
P2
372
372A
FRONT
LEFT
RIGHT
REAR

FIG. 22



71
49
141
84
C1
P1
99
91
21
81A
92
71A
P2
93
86
145
V1
V2
94
85
87
83
372
372A
71B
81
143
AX3
43
41
70
144
109
20
40
142
23
140
300A
FRONT
LEFT
RIGHT
REAR

FIG. 23



40
70
71
41
143
109
713
86
372
272
141
272A
711
372A
84
99
71B
91
92
83
71A
94
145
87
85
712
72
142
80
82
140
60
30
62
50
130
20
52
132
134
144
300B
DOWN
LEFT
FRONT
REAR
RIGHT
UP

CASSETTE INCLUDING PROTRUSION CONFIGURED TO CONTACT WIDTHWISE CENTER PORTION OF INK RIBBON TO CAUSE INK RIBBON TO BE CURVED

REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority from Japanese Patent Application No. 2024-034251 filed on Mar. 6, 2024. The entire content of the priority application is incorporated herein by reference.

BACKGROUND ART

[0002] Cassettes including ink ribbons for use in printers are known. For example, a conventional cassette includes a first frame part, a second frame part, an auxiliary tape roll, and a take-up spool. The second frame part accommodates therein the auxiliary tape roll and the take-up spool. The auxiliary tape roll is configured by winding an ink ribbon into a roll. The take-up spool is configured to take up the ink ribbon from the auxiliary tape roll.

[0003] The second frame part includes a front wall, a rear wall, a left wall, and a right wall. The front wall and the rear wall are arranged in a front-rear direction. The left wall and the right wall are arranged in a left-right direction. The second frame part further includes a first inner wall, a second inner wall, a third inner wall, and a fourth inner wall. The first inner wall and the second inner wall are arranged in this order from front to rear at positions between the front wall and the rear wall. The third inner wall and the fourth inner wall are arranged in this order from left to right at positions between the left wall and the right wall.

[0004] The third inner wall has a front edge that is connected to a left edge of the first inner wall. The third inner wall also has a rear edge that is connected to a left edge of the second inner wall. The fourth inner wall has a rear edge that is connected to a right edge of the second inner wall.

[0005] The second frame part is formed with an exposed area. The exposed area extends from a space between a right edge of the front wall and a right edge of the first inner wall to a space between a front edge of the fourth inner wall and a front edge of the right wall. The ink ribbon is drawn off the auxiliary tape roll, passes sequentially through a portion between the left wall and the third inner wall, a portion between the front wall and the first inner wall, the exposed area, and a portion between the fourth inner wall and the right wall, and is taken up by the take-up spool.

SUMMARY

[0006] In the conventional cassette described above, the ink ribbon may become skewed due to various factors such as part variation, thereby affecting the printing quality produced with the ink ribbon.

[0007] In view of the foregoing, it is an object of the present disclosure to provide a cassette that can suppress skewing of an ink ribbon.

[0008] In order to attain the above and other object, the present disclosure provides a cassette including: a lower case; a ribbon roll; and a take-up spool. The lower case has a box-like shape. The ribbon roll is accommodated in the lower case. The ribbon roll is a roll of an ink ribbon. The ribbon roll is rotatable about a first axis extending in an up-down direction. The take-up spool is accommodated in the lower case. The take-up spool is configured to take up the ink ribbon from the ribbon roll. The lower case includes: a front wall; a rear wall; a left wall; a right wall; a first inner wall; a second inner wall; a third inner wall; and a fourth inner wall. The front wall extends in both the up-down direction and a left-right direction orthogonal to the up-down direction. The front wall has a right edge. The rear wall extends in both the up-down direction and the left-right direction. The front wall and the rear wall are arranged in a front-rear direction orthogonal to both the up-down direction and the left-right direction. The left wall extends in both the up-down direction and the front-rear direction. The right wall extends in both the up-down direction and the front-rear direction. The left wall and the right wall are arranged in the left-right direction. The right wall has a front edge. The first inner wall extends in both the up-down direction and the left-right direction. The first inner wall is positioned between the front wall and the rear wall. The first inner wall has a left edge and a right edge. The second inner wall extends in both the up-down direction and the left-right direction. The second inner wall is positioned between the front wall and the rear wall and positioned further rearward relative to the first inner wall in the front-rear direction. The second inner wall has a left edge and a right edge. The third inner wall extends in both the up-down direction and the front-rear direction. The third inner wall is positioned between the left wall and the right wall. The third inner wall has a front edge connected to the left edge of the first inner wall and a rear edge connected to the left edge of the second inner wall. The fourth inner wall extends in both the up-down direction and the front-rear direction. The fourth inner wall is positioned between the left wall and the right wall and positioned further rightward relative to the third inner wall in the left-right direction. The fourth inner wall has a front edge and a rear edge connected to the right edge of the second inner wall. The lower case has an exposed area in which the ink ribbon is exposed. The exposed area extends from a portion between the right edge of the front wall and the right edge of the first inner wall to a portion between the front edge of the fourth inner wall and the front edge of the right wall. The lower case defines a conveying path along which the ink ribbon is conveyed in a conveying direction when the take-up spool takes up the ink ribbon from the ribbon roll. The conveying path includes a first part, a second part, the exposed area, and a third part from an upstream side in the conveying direction. The first part passes between the left wall and the third inner wall. The second part passes between the front wall and the first inner wall. The third part passes between the fourth inner wall and the right wall. The cassette further includes: a protrusion protruding rightward from a portion of the fourth inner wall other than the front edge and the rear edge of the fourth inner wall. The protrusion is configured to contact a widthwise center portion of the ink ribbon to cause the ink ribbon to be curved along a width of the ink ribbon so that the widthwise center portion of the ink ribbon protrudes rightward.

[0009] In the above structure, the protrusion causes the ink ribbon to be curved along the width of the ink ribbon so that the widthwise center portion of the ink ribbon protrudes rightward. Accordingly, in a case where the ink ribbon attempts to be displaced along the width thereof, a force directed to a direction for restricting the displacement of the ink ribbon is applied to the ink ribbon. Therefore, the cassette according to the above aspect can restrain skewing of the ink ribbon.

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. **1** is a perspective view of a printer **1** and a cassette **100**.

[0011] FIG. **2** is a plan view of an attachment portion **11** to which the cassette **100** is not attached.

[0012] FIG. **3** is a plan view of the attachment portion **11** to which the cassette **100** has been attached.

[0013] FIG. **4** is a perspective view of the cassette **100**.

[0014] FIG. **5** is an exploded perspective view of the cassette **100**.

[0015] FIG. **6** is a cross-sectional view taken along line VI-VI in FIG. **10**.

[0016] FIG. **7** is a perspective view of the cassette **100** from which a lower cover **70** has been removed.

[0017] FIG. **8** is another perspective view of the cassette **100** from which the lower cover **70** has been removed.

[0018] FIG. **9** is a perspective view of the lower cover **70**.

[0019] FIG. **10** is a bottom view of the cassette **100**.

[0020] FIG. **11** is a bottom view of the cassette **100** from which the lower cover **70** has been removed.

[0021] FIG. **12** is a cross-sectional view taken along line XII-XII in FIG. **11**.

[0022] FIG. **13** is an enlarged view of a region W**1** illustrated in FIG. **11**.

[0023] FIG. **14**A is an explanatory view illustrating the way how a printing tape **101** is fixed to a tape spool **102**.

[0024] FIG. **14**B is another explanatory view illustrating the way how the printing tape **101** is fixed to the tape spool **102**.

[0025] FIG. **15**A is a schematic cross-sectional view taken along a plane containing an axis AX**3**, and illustrating a configuration in which a first gap G**1** has a size the same as a size of a second gap G**2**.

[0026] FIG. **15**B is a schematic cross-sectional view taken along a plane containing the axis AX**3**, and illustrating a configuration in which the size of the first gap G**1** is greater than the size of the second gap G**2**.

[0027] FIG. **15**C is a schematic cross-sectional view taken along a plane containing the axis AX**3**, and illustrating a configuration in which the size of the first gap G**1** is greater than the size of the second gap G**2** and the axis AX**3** is inclined relative to an up-down direction.

[0028] FIG. **16** is a partial perspective view of a cassette **100**A, and particularly illustrating a portion in the vicinity of a head front peripheral wall **184**.

[0029] FIG. **17** is a partial perspective view of a cassette **100**B, and particularly illustrating a portion in the vicinity of a head front peripheral wall **284**.

[0030] FIG. **18** is a perspective view of a cassette **200**.

[0031] FIG. **19** is a bottom view of the cassette **200**.

[0032] FIG. **20** is a perspective view of a cassette **300** from which a lower cover **70** has been removed.

[0033] FIG. **21** is a bottom view of the cassette **300** from which the lower cover **70** has been removed.

[0034] FIG. **22** is a bottom view of a cassette **300**A.

[0035] FIG. **23** is a perspective view of a cassette **300**B.

DESCRIPTION

[0036] Hereinafter, embodiments of the present disclosure will be described while referring to the accompanying drawings. Note that the drawings are used to describe the technical features made possible with the present disclosure. In other words, the configurations and the like illustrated in the drawings are merely illustrative examples, and the present disclosure is not intended to be limited to these configurations.

First Embodiment

[0037] First, a cassette **100** according to a first embodiment will be described with reference to FIGS. **1** through **15**. The cassette **100** includes an ink ribbon **105** and a printing tape **101**. The cassette **100** is used in a printer **1**. The printing tape **101** has a strip-like shape and is a type of printing medium. The ink ribbon **105** has a strip-like shape, and has a surface coated with ink. In the following description, the lower-left side, the upper-right side, the upper-left side, the lower-right side, the lower side, and the upper side in FIG. **1** will be referred to as the right side, the left side, the front side, the rear side, the lower side, and the upper side of the cassette **100** and the printer **1**, respectively.

[0038] Next, the printer **1** will be described with reference to FIGS. **1** through **3**. The printer **1** is a thermal printer configured to perform printing on the printing tape **101** using the ink ribbon **105**. The printer **1** includes a housing **2** and a cover **3** (see FIG. **1**). The housing **2** has a rectangular parallelepiped shape. An attachment portion **11** is provided in an upper surface of the housing **2** on a right portion thereof. The attachment portion **11** is a recess that is recessed downward from the upper surface of the housing **2**. The attachment portion **11** has a shape that conforms to an outer shape of the cassette **100** in a plan view. The cassette **100** is attachable to the attachment portion **11** (see FIG. **3**). A discharge opening **12** is formed in a right surface of the housing **2** for discharging, from the housing **2**, the printing tape **101** on which printing has been performed. The discharge opening **12** has a slit-like shape and extends in an up-down direction. The cover **3** has a plate shape and extends in both a front-rear direction and a left-right direction. The cover **3** is attachable to the housing **2** from above. In a state where the cover **3** is attached to the housing **2**, the cover **3** covers the attachment portion **11** from above. The cover **3** is removable from the housing **2**. In a state where the cover **3** is removed from the housing **2**, the attachment portion **11** is open upward. Note that the up-down direction, the front-rear direction, and the left-right direction are orthogonal to each other.

[0039] The printer **1** also includes a thermal head **4**, a platen unit **5**, and a rotational shaft **6** which are provided in the attachment portion **11**. The thermal head **4** is provided in a right portion in the attachment portion **11**, and extends upward from a bottom surface defining the attachment portion **11**. The thermal head **4** has a plate-like shape, and extends in both the left-right direction and the up-down direction. The thermal head **4** includes a plurality of heating elements. The heating elements are arranged in the up-down direction on a front surface of the thermal head **4**. The selected one or more heating elements of the thermal head **4** are heated, thereby perform printing on the printing tape **101**.

[0040] The platen unit **5** includes a platen holder **15**, a pivot shaft **16**, a platen shaft **17**, a platen roller **18**, and a platen gear **19**. The platen holder **15** is disposed in the attachment portion **11** at a position frontward of the thermal head **4**. The platen holder **15** extends in the left-right direction, and has an open end open rearward.

[0041] The pivot shaft **16** is disposed on a left end of the platen holder **15**, and extends in the up-down direction. The pivot shaft **16** supports the platen holder **15** so that the platen holder **15** is pivotally movable. A lower end of the pivot shaft **16** is fixed to the housing **2**. In a plan view, the platen holder **15** is pivotally movable about the pivot shaft **16** so that a right end of the platen holder **15** is moved in the front-rear direction (see FIGS. **2** and **3**). The right end of the platen holder **15** is urged by a spring (not illustrated) so that the platen older **15** is pivotally moved frontward about the pivot shaft **16**.

[0042] The platen shaft **17** is rotatably supported on the right end of the platen holder **15**, and extends in the up-down direction. The platen roller **18** extends in the up-down direction and is attached to a lower portion of the platen shaft **17**. The platen gear **19** is attached to an upper end of the platen shaft **17**.

[0043] The rotational shaft **6** is disposed in the attachment portion **11** at a position rearward of the thermal head **4**. The rotational shaft **6** extends in the up-down direction. The rotational shaft **6** has a lower end supported by the bottom surface of the attachment portion **11** so that the rotational shaft **6** is rotatable. A conveying motor (not illustrated) is disposed at a position rearward of the attachment portion **11**. A driving force of the conveying motor is transmitted to the rotational shaft **6** via a gear train (not illustrated).

[0044] Next, the cassette **100** will be described with reference to FIGS. **4** through **15**. As illustrated in FIGS. **4** and **5**, the cassette **100** includes a case **20**. The case **20** includes a lower case **40** and an upper case **30**, and is configured by assembling the upper case **30** with the lower case **40** from above. The lower case **40** constitutes a lower portion of the case **20**. The lower case **40** has a box-like shape, i.e., a rectangular parallelepiped shape with a lower-right corner portion thereof being cut out.

[0045] A head opening **99** is formed in the lower case **40**. The head opening **99** is a recess that is first recessed rearward from a front-right corner of the lower case **40** and is then recessed leftward. The head opening **99** is open downward and diagonally rightward and frontward. Hereinafter, a portion of the lower case **40** which is positioned further frontward relative to the head opening **99** will be referred to as an arm portion **49**. The arm portion **49** extends rightward from a left-front corner of the lower case **40**. The upper case **30** is positioned further upward relative to the lower case **40**, and constitutes an upper portion of the case **20**. The upper case **30** has a box-like shape, i.e., rectangular parallelepiped shape.

[0046] As illustrated in FIGS. **5** through **8**, the cassette **100** also includes a ribbon roll **107**, a take-up spool **109**, and a tape roll **103** (see FIG. **5**). The ribbon roll **107** is accommodated in the lower case **40**. The ribbon roll **107** is configured by winding the ink ribbon **105** around a ribbon spool **106**. That is, the ribbon roll **107** is a roll of the ink ribbon **105**. The ribbon spool **106** has a hollow cylindrical shape that is elongated in the up-down direction. The take-up spool **109** is accommodated in the lower case **40**. The take-up spool **109** is configured to take up the ink ribbon **105** from the ribbon roll **107**.

[0047] The take-up spool **109** has a cylindrical shape (specifically, a hollow cylindrical shape) that is elongated in the up-down direction. In the present embodiment, a lower end of the take-up spool **109** has an outer diameter R**11** that is greater than an inner diameter R**12** of an upper end of the take-up spool **109** (see FIG. **6**). The take-up spool **109** includes guide discs **109**A and **109**B. The guide disc **109**A is provided at a position just upward of the lower end of the take-up spool **109**, and protrudes radially outward from an outer circumferential surface of the take-up spool **109**. The guide disc **109**B is provided at a position just downward of the upper end of the take-up spool **109**, and protrudes radially outward from the outer circumferential surface of the take-up spool **109**. The guide discs **109**A and **109**B are configured to guide the ink ribbon **105**, which is wound around the take-up spool **109** between the guide discs **109**A and **109**B, to restrain the ink ribbon **105** from moving in the up-down direction.

[0048] In the present embodiment, the lower end of the take-up spool **109** is a portion of the take-up spool **109** that is further downward relative to the guide disc **109**A. The upper end of the take-up spool **109** is a portion of the take-up spool **109** that is further upward relative to the guide disc **109**B. A clutch spring **119** (see FIG. **6**) is wound around an outer circumference of the upper end of the take-up spool **109**. The clutch spring **119** exerts rotational resistance on the take-up spool **109** when the take-up spool **109** takes up the ink ribbon **105** from the ribbon roll **107**.

[0049] As illustrated in FIGS. **5** and **6**, the tape roll **103** is accommodated in the upper case **30**. The tape roll **103** is configured by winding the printing tape **101** around a tape spool **102**. That is, the tape roll **103** is a roll of the printing tape **101**. The tape spool **102** has a hollow cylindrical shape and extends in the up-down direction.

[0050] Next, the upper case **30** will be described with reference to FIGS. **4** and **5**. As illustrated in FIGS. **4** and **5**, the upper case **30** includes an upper wall **51**, a support wall **61** (see FIG. **5**), and an outer peripheral wall **130**. The upper wall **51** constitutes an upper edge of the upper case **30**. The upper wall **51** extends in both the front-rear direction and the left-right direction, and has a rectangular shape in a plan view. As illustrated in FIG. **5**, the support wall **61** is arranged at a position further downward relative to the upper wall **51** and further upward relative to a lower edge of the upper case **30**. The support wall **61** extends in both the front-rear direction and the left-right direction, and has a rectangular shape in a plan view. The upper wall **51** and the support wall **61** are arranged in the up-down direction.

[0051] The outer peripheral wall **130** extends from a peripheral edge of the upper wall **51** through a peripheral edge of the support wall **61** to a position further downward relative to the peripheral edge of the support wall **61**. The outer peripheral wall **130** includes a front wall **131**, a rear wall **132**, a left wall **133**, and a right wall **134**. The front wall **131** constitutes a front edge of the upper case **30**, and extends in both the left-right direction and the up-down direction. The rear wall **132** constitutes a rear edge of the upper case **30**, and extends in both the left-right direction and the up-down direction. The front wall **131** and the rear wall **132** are arranged in the front-rear direction.

[0052] The left wall **133** constitutes a left edge of the upper case **30**, and extends in both the front-rear direction and the up-down direction. A front edge of the left wall **133** is connected to a left edge of the front wall **131**. A rear edge of the left wall **133** is connected to a left edge of the rear wall **132**. The right wall **134** constitutes a right edge of the upper case **30**, and extends in both the front-rear direction and the up-down direction. The left wall **133** and the right wall **134** are arranged in the left-right direction. A front edge of the right wall **134** is connected to a right edge of the front wall **131**. A rear edge of the right wall **134** is connected to a right edge of the rear wall **132**.

[0053] As illustrated in FIG. **5**, a support shaft **31** is provided on the upper wall **51**. The support shaft **31** is disposed on a center of the upper wall **51** with respect to the front-rear direction and the left-right direction, and protrudes downward from a lower surface of the upper wall **51**. The support shaft **31** supports the tape roll **103** in cooperation with a support shaft **48** described later (see FIG. **9**). The tape roll **103** is rotatable about an axis AX**2**. The axis AX**2** extends in the up-down direction and passes through a center of the support shaft **31**.

[0054] Three support shafts **32**, **33**, and **34** are provided on the support wall **61**. The support shaft **32** is disposed on a front-right portion of the support wall **61**, and protrudes downward from a lower surface of the support wall **61**. The support shaft **32** supports an output gear **21** so that the output gear **21** is rotatable. The support shaft **33** is disposed at a position diagonally leftward and rearward of the support shaft **32**, and protrudes downward from the lower surface of the support wall **61**. The support shaft **33** supports a coupling gear **22** so that the coupling gear **22** is rotatable. A right-front edge of the coupling gear **22** is in meshing engagement with a left-rear edge of the output gear **21**. The support shaft **34** is disposed at a position diagonally rightward and rearward of the support shaft **33**, and protrudes downward from the lower surface of the support wall **61**. The support shaft **34** supports an input gear **23** so that the input gear **23** is rotatable. A left-front edge of the input gear **23** is in meshing engagement with a right-rear edge of the coupling gear **22**.

[0055] A tape guide slit (not illustrated) is formed in the support wall **61**. The tape guide slit is formed in a rear portion of the support wall **61** for guiding the printing tape **101** from the upper case **30** to a tape insertion opening **44** (described later). The tape guide slit penetrates the support wall **61** in the up-down direction, and extends in the left-right direction.

[0056] As illustrated in FIGS. **4** through **6**, the upper case **30** includes an upper cover **50** and a lower cover **60**, and is configured by assembling the upper cover **50** with the lower cover **60** from above. The upper cover **50** constitutes an upper portion of the upper case **30**, and includes the upper wall **51** and an outer peripheral wall **52**. The outer peripheral wall **52** extends downward from the peripheral edge of the upper wall **51**. The lower cover **60** constitutes a lower portion of the upper case **30**, and includes the support wall **61** illustrated in FIG. **5** and an outer peripheral wall **62**. The outer peripheral wall **62** extends both upward and downward from the peripheral edge of the support wall **61**.

[0057] The upper cover **50** is assembled with the lower cover **60** by engaging the outer peripheral wall **52** with the outer peripheral wall **62** from above. The outer peripheral wall **52** and the outer peripheral wall **62** assembled in this way configure the outer peripheral wall **130**. Here, "engaging the outer peripheral wall **52** with the outer peripheral wall **62**" signifies that one or more protrusions provided on one of the outer peripheral wall **52** and the outer peripheral wall **62** are respectively fitted in one or more recesses formed in the other of the outer peripheral wall **52** and the outer peripheral wall **62**, or that one or more hooks provided on one of the outer peripheral wall **52** and the outer peripheral wall **62** are respectively hooked in one or more engagement holes formed in the other of the outer peripheral wall **52** and the outer peripheral wall **62**.

[0058] Next, the lower case **40** will be described with reference to FIGS. **4** through **10**. As illustrated in FIGS. **4** through **6**, the lower case **40** includes a lower wall **71**, an intermediate wall **81**, an outer peripheral wall **140**, a head peripheral wall **83**, and a partitioning wall **88**. The lower wall **71** constitutes a lower edge of the lower case **40**. The lower wall **71** extends in both the front-rear direction and the left-right direction, and has a rectangular shape in a plan view. A front edge of the lower wall **71** configures a lower wall of the arm portion **49**. As illustrated in FIG. **5**, the intermediate wall **81** is arranged further upward relative to the lower wall **71** and constitutes an upper edge of the lower case **40**. The intermediate wall **81** extends in both the front-rear direction and the left-right direction, and has a rectangular shape in a plan view. A front edge of the intermediate wall **81** configures an upper wall of the arm portion **49**. The lower wall **71** and the intermediate wall **81** are arranged in the up-down direction.

[0059] The outer peripheral wall **140** extends from a peripheral edge of the lower wall **71** to a peripheral edge of the intermediate wall **81**. That is, the outer peripheral wall **140** extends between the lower wall **71** and the intermediate wall **81**. The outer peripheral wall **140** includes a front wall **141**, a rear wall **142**, a left wall **143**, and a right wall **144**. The front wall **141** constitutes a front edge of the lower case **40**, and extends in both the left-right direction and the up-down direction. A right edge of the front wall **141** is in the approximate center of the lower case **40** in the left-right direction. The front wall **141** configures a front wall of the arm portion **49**. The rear wall **142** constitutes a rear edge the lower case **40**, and extends in both the left-right direction and the up-down direction. The front wall **141** and the rear wall **142** are arrange in the front-rear direction.

[0060] The left wall **143** constitutes a left edge of the lower case **40**, and extends in both the front-rear direction and the up-down direction. A front edge of the left wall **143** is connected to a left edge of the front wall **141**. A rear edge of the left wall **143** is connected to a left edge of the rear wall **142**. The right wall **144** constitutes a right edge of the lower case **40**, and extends in both the front-rear direction and the up-down direction. The left wall **143** and the right wall **144** are arranged in the left-right direction. A rear edge of the right wall **144** is connected to a right edge of the rear wall **142**.

[0061] As illustrated in FIGS. **7** and **8**, the head peripheral wall **83** extends downward from a front-right portion of the intermediate wall **81**. An upper edge of the head peripheral wall **83** is connected to the intermediate wall **81**. The head peripheral wall **83** surrounds the head opening **99**. Specifically, the head peripheral wall **83** includes a head front peripheral wall **84**, a head rear peripheral wall **85**, a head left peripheral wall **86**, and a head right peripheral wall **87**.

[0062] Each of the head front peripheral wall **84** and the head rear peripheral wall **85** extends in both the left-right direction and the up-down direction, and is arranged between the front wall **141** and the rear wall **142**. The head front peripheral wall **84** and the head rear peripheral wall **85** are arranged from front to rear in this order. That is, the head rear peripheral wall **85** is positioned further rearward relative to the head front peripheral wall **84**. The head front peripheral wall **84** configures a rear wall of the arm portion **49**.

[0063] Each of the head left peripheral wall **86** and the head right peripheral wall **87** extends in both the front-rear direction and the up-down direction, and is arranged between the left wall **143** and the right wall **144**. The head left peripheral wall **86** and the head right peripheral wall **87** are arranged from left to right in this order. That is, the head right peripheral wall **87** is positioned further rightward relative to the head left peripheral wall **86**.

[0064] A right edge of the head front peripheral wall **84** is located at approximately the same position in the left-right direction as the right edge of the front wall **141**. A left edge of the head front peripheral wall **84** is positioned further rightward relative to the left wall **143**. A right edge of the head rear peripheral wall **85** is positioned further leftward relative to the right wall **144**. A left edge of the head rear peripheral wall **85** is located at the same position in the left-right direction as the left edge of the head front peripheral wall **84**. A front edge of the head left peripheral wall **86** is connected to the left edge of the head front peripheral wall **84**. A rear edge of the head left peripheral wall **86** is connected to the left edge of the head rear peripheral wall **85**. A rear edge of the head right peripheral wall **87** is connected to the right edge of the head rear peripheral wall **85**. A front edge of the head right peripheral wall **87** is positioned further rearward relative to the front wall **141**.

[0065] A length of the head front peripheral wall **84** in the left-right direction is smaller than a length of the head rear peripheral wall **85** in the left-right direction. Lengths of the head left peripheral wall **86** and the head right peripheral wall **87** in the front-rear direction are smaller than the length of the head rear peripheral wall **85** in the left-right direction and smaller than the length of the head front peripheral wall **84** in the left-right direction.

[0066] The head front peripheral wall **84** includes path-defining parts **98**A and **98**B, and a main portion **84**A. The path-defining part **98**A is provided on the left edge of the head front peripheral wall **84**. The path-defining part **98**A is a solid cylindrical member extending in the up-down direction and has a pin shape. During the conveyance of the ink ribbon **105**, the ink ribbon **105** contacts the path-defining part **98**A in an entirety of the width (in an entirety of a dimension in the up-down direction) of the ink ribbon **105** to have a contact portion with the path-defining part **98**A, whereby a direction of a path of the ink ribbon **105** is changed. Specifically, the path of the ink ribbon **105** curves around an outer circumference of the path-defining part **98**A. The contact portion between the ink ribbon **105** and the path-defining part **98**A has a straight-line shape extending in the up-down direction when viewed in a longitudinal direction of the ink ribbon **105**.

[0067] The path-defining part **98**B is provided on the right edge of the head front peripheral wall **84**. The path-defining part **98**B is a solid cylindrical member that extends in the up-down direction and has a pin shape. During the conveyance of the ink ribbon **105**, the ink ribbon **105** contacts the path-defining part **98**B in an entirety of the width (in an entirety of a dimension in the up-down direction) of the ink ribbon **105** to have a contact portion with the path-defining part **98**B, whereby a direction of the path of the ink ribbon **105** is changed. Specifically, the path of the ink ribbon **105** curves around an outer circumference of the path-defining part **98**B. The contact portion between the ink ribbon **105** and the path-defining part **98**B has a straight-line shape that extends in the up-down direction when viewed in the longitudinal direction of the ink ribbon **105**.

[0068] The main portion **84**A is a portion of the head front peripheral wall **84** other than both the path-defining part **98**A and the path-defining part **98**B. That is, the main portion **84**A is a portion of the head front peripheral wall **84** between the path-defining part **98**A and the path-defining part **98**B. The main portion **84**A has a plate shape and extends in both the left-right direction and the up-down direction. The outer circumferences of the path-defining part **98**A and the path-defining part **98**B protrude frontward from the main portion **84**A.

[0069] The partitioning wall **88** extends in both the left-right direction and the up-down direction, and is arranged between the front wall **141** and the head front peripheral wall **84**. An upper edge of the partitioning wall **88** is connected to the intermediate wall **81**. A right edge of the partitioning wall **88** is located at approximately the same position in the left-right direction as the right edge of the head front peripheral wall **84**. The partitioning wall **88** partitions a space between the front wall **141** and the head front peripheral wall **84** into a path for the ink ribbon **105** and a path for the printing tape **101**. Specifically, a space between the partitioning wall **88** and the head front peripheral wall **84** is the path for the ink ribbon **105**, and a space between the partitioning wall **88** and the front wall **141** is the path for the printing tape **101**.

[0070] The intermediate wall **81** includes an upper bridging portion **81**A. The upper bridging portion **81**A is a portion of the intermediate wall **81** spanning between an upper edge of the head front peripheral wall **84** and an upper edge of the head rear peripheral wall **85** in the front-rear direction. That is, the upper bridging portion **81**A connects the upper edge of the head front peripheral wall **84** and the upper edge of the head rear peripheral wall **85** to each other in the front-rear direction. Therefore, even if a force directed frontward were to act on the head front peripheral wall **84**, for example, the upper bridging portion **81**A would not contract substantially in the front-rear direction, thereby restraining the upper edge of the head front peripheral wall **84** from approaching the head rear peripheral wall **85** substantially in the front-rear direction.

[0071] As illustrated in FIGS. **5** through **8**, a support shaft **42** and an upper support portion **43** are provided on the intermediate wall **81**. Also, the tape insertion opening **44** (see FIGS. **7** and **8**) is formed in the intermediate wall **81**. The support shaft **42** is disposed on a left portion of the intermediate wall **81** and protrudes downward from a lower surface of the intermediate wall **81**. The support shaft **42** supports the ribbon roll **107**. Thus, the ribbon roll **107** is disposed between the head rear peripheral wall **85** and the rear wall **142** and between the left wall **143** and the right wall **144** (specifically, between head right peripheral wall **87** and the right wall **144**). The ribbon roll **107** is rotatable about an axis AX**1** (see FIG. **5**). The axis AX**1** extends in the up-down direction and passes through the center of the support shaft **42**.

[0072] The upper support portion **43** is formed on the intermediate wall **81** at a position rightward of the support shaft **42**. The upper support portion **43** is an outer circumferential wall having an annular shape and protrudes slightly downward from the intermediate wall **81**. The upper support portion **43** is fitted in the take-up spool **109** from above. In other words, the upper support portion **43** is disposed inside an inner circumference of the take-up spool **109** in a radial direction of the take-up spool **109**. The upper support portion **43** supports an upper end of the take-up spool **109** so that the take-up spool **109** is rotatable. A shaft insertion hole that penetrates the intermediate wall **81** in the up-down direction is provided on an inner side of the upper support portion **43**.

[0073] As illustrated in FIGS. **7** and **8**, the tape insertion opening **44** is formed in a left-rear corner of the intermediate wall **81**. The tape insertion opening **44** is an opening that penetrates the intermediate wall **81** in the up-down direction. The printing tape **101** is configured to be inserted into the lower case **40** through the tape insertion opening **44** from above.

[0074] As illustrated in FIGS. **9** and **10**, a first notch **71**A and a second notch **71**B are formed in the lower wall **71**. The first notch **71**A extends rearward from a right-front corner of the lower wall **71**. Specifically, the first notch **71**A is cut out rearward from a portion of the front edge of the lower wall **71** that is further rightward relative the right edge of the front wall **141** to a portion between the head front peripheral wall **84** and the head rear peripheral wall **85** in the front-rear direction. The second notch **71**B extends leftward from a rear end of the first notch **71**A. Specifically, the second notch **71**B is formed continuously with the first notch **71**A and is cut out in the left-right direction between the head front peripheral wall **84** and the head rear peripheral wall **85**.

[0075] An edge of the lower wall **71** forming a left edge of the first notch **71**A defines a lower edge of a guide opening **91** (described later). An edge of the lower wall **71** forming a right edge of the first notch **71**A is connected to a lower edge of the head right peripheral wall **87**. An edge of the lower wall **71** forming a front edge of the second notch **71**B is connected to a lower edge of the head front peripheral wall **84**. An edge of the lower wall **71** forming a rear edge of the second notch **71**B is connected to a lower edge of the head rear peripheral wall **85**. An edge of the lower wall **71** forming a left edge of the second notch **71**B is connected to a lower edge of the head left peripheral wall **86**. Accordingly, the first notch **71**A and the second notch **71**B constitute a lower end of the head opening **99**.

[0076] A lower support portion **41** and the support shaft **48** (see FIG. **9**) are provided on the lower wall **71**. The lower support portion **41** is provided in a right-rear portion of the lower wall **71**. The lower support portion **41** is an inner circumferential wall defining a support hole that penetrates the lower wall **71** in the up-down direction.

[0077] As illustrated in FIG. **6**, the take-up spool **109** is fitted in the lower support portion **41**. That is, the lower support portion **41** is disposed outside an outer circumference of the take-up spool **109** in the radial direction of the take-up spool **109**. The lower support portion **41** supports a lower end of the take-up spool **109** so that the take-up spool **109** is rotatable. In other words, the take-up spool **109** is supported from above by the upper support portion **43** and from below by the lower support portion **41**. The take-up spool **109** is rotatable about an axis AX**3** (see FIGS. **5** and **10**). The axis AX**3** extends in the up-down direction and passes through centers of the lower support portion **41** and the upper support portion **43**. The axis AX**3** is a diametrical center of the take-up spool **109** in a plan view. The take-up spool **109** is arranged between the head rear peripheral wall **85** and the rear wall **142** and between the left wall **143** and the right wall **144** (specifically, between head right peripheral wall **87** and the right wall **144**).

[0078] In the radial direction of the take-up spool **109**, a first gap G**1** is formed between the lower end of the take-up spool **109** and the lower support portion **41**, and a second gap G**2** is formed between the upper end of the take-up spool **109** and the upper support portion **43**, as illustrated in FIG. **6**. In other words, the take-up spool **109** is supported by the lower support portion **41** with the first gap G**1** present between the lower end of the take-up spool **109** and the lower support portion **41** in the radial direction, and is supported by the upper support portion **43** with the second gap G**2** present between the upper end of the take-up spool **109** and the upper support portion **43** in the radial direction.

[0079] The first gap G**1** has a size that is equivalent to a radius of the outer diameter R**11** at the lower end of the take-up spool **109** minus a radius of an inner diameter of the lower support portion **41**. While not limited to any specific size, the first gap G**1** is approximately 0.3 mm in the present embodiment. The second gap G**2** has a size that is equivalent to a radius of an outer diameter of the upper support portion **43** minus a radius of the inner diameter R**12** at the upper end of the take-up spool **109**. While not limited to any specific size, the second gap G**2** is approximately 0.06 mm in the present embodiment.

[0080] That is, the first gap G**1** is different in size from the second gap G**2**, and is greater than the second gap G**2** in the present embodiment. An absolute value of the difference between the first gap G**1** and the second gap G**2** (|G**1**−G**2**|) is less than a thickness T**3** of the lower support portion **41** in the up-down direction and is less than a thickness T**4** of the upper support portion **43** in the up-down direction. In the present embodiment, the absolute value of the difference between the first gap G**1** and the second gap G**2** is approximately 0.24 mm.

[0081] As illustrated in FIG. **9**, the support shaft **48** is located diagonally leftward and frontward of the lower support portion **41**, and protrudes upward from an upper surface of the lower wall **71**. The support shaft **48** opposes the support shaft **31** in the up-down direction (see FIG. **5**). The support shaft **48** supports the tape roll **103**.

[0082] As illustrated in FIGS. **9** and **10**, the lower wall **71** includes an opening edge **46**. The opening edge **46** is provided in a portion of the lower wall **71** that is further frontward relative to the second notch **71**B, i.e., in the arm portion **49**. The opening edge **46** has a rectangular shape whose dimension in the left-right direction is greater than that in the front-rear direction, and defines an engagement hole **45**. The engagement hole **45** penetrates the lower wall **71** in the up-down direction and is elongated in the left-right direction. A lower end **88**A of the partitioning wall **88** is fitted in the engagement hole **45** (see FIG. **10**).

[0083] The lower wall **71** includes a rib **46**A. The rib **46**A is disposed in a portion of the opening edge **46**A defining a right edge of the engagement hole **45**. Specifically, the rib **46**A protrudes rearward from a front edge of the opening edge **46**, i.e., into the engagement hole **45**. In other words, the rib **46**A protrudes in the front-rear direction from the opening edge **46** toward the lower end **88**A of the partitioning wall **88**. The rib **46**A has a rear edge that is either slightly separated from a front edge of the lower end **88**A of the partitioning wall **88** or contacts the lower end **88**A of the partitioning wall **88**. The rib **46**A is at least positioned sufficiently close to the lower end **88**A of the partitioning wall **88** so that the rear edge of the rib **46**A contacts the lower end **88**A of the partitioning wall **88** when a force directed rearward is applied to the arm portion **49**.

[0084] As illustrated in FIGS. **4** and **5**, the lower case **40** includes an upper cover **80** and a lower cover **70**, and is configured by assembling the upper cover **80** with the lower cover **70** from above. The upper cover **80** constitutes an upper portion of the lower case **40**. The upper cover **80** includes the intermediate wall **81**, an outer peripheral wall **82**, the head peripheral wall **83**, and the partitioning wall **88**. The outer peripheral wall **82** extends downward from the peripheral edge of the intermediate wall **81**. The lower cover **70** constitutes a lower portion of the lower case **40**. The lower cover **70** includes the lower wall **71**, and an outer peripheral wall **72**. The outer peripheral wall **72** extends upward from the peripheral edge of the lower wall **71**.

[0085] The lower cover **70** is assembled with the upper cover **80** by engaging the outer peripheral wall **72** with the outer peripheral wall **82** from below. Through this assembly, the outer peripheral wall **72** and the outer peripheral wall **82** configure the outer peripheral wall **140**. Here, "engaging the outer peripheral wall **72** with the outer peripheral wall **82**" signifies that one or more protrusions provided on one of the outer peripheral wall **72** and the outer peripheral wall **82** are fitted in one or more recesses formed in the other of the outer peripheral wall **72** and the outer peripheral wall **82**, or that one or more hooks provided on one of the outer peripheral wall **72** and the outer peripheral wall **82** are hooked in one or more engagement holes formed in the other of the outer peripheral wall **72** and the outer peripheral wall **82**.

[0086] As illustrated in FIGS. **4** and **7**, a guide opening **91**, an exposed area **92**, an entry opening **93**, and a discharge opening **94** are formed in the lower case **40**. The guide opening **91** is formed between the right edge of the head front peripheral wall **84** and the right edge of the front wall **141**. The guide opening **91** has a slit shape that extends in the up-down direction. The guide opening **91** functions to guide the printing tape **101** and the ink ribbon **105** from the interior of the arm portion **49** into the exposed area **92**, as illustrated in FIG. **4**.

[0087] The exposed area **92** constitutes a right-front area of the head opening **99**, and extends from a portion between the right edge of the front wall **141** and the right edge of the head front peripheral wall **84** to a portion between the front edge of the head right peripheral wall **87** and the front edge of the right wall **144**. The exposed area **92** allows a portion of the printing tape **101** and the ink ribbon **105** positioned between the guide opening **91** and the discharge opening **94** to be exposed outside the lower case **40**, as illustrated in FIG. **4**.

[0088] The entry opening **93** is formed between the front edge of the head right peripheral wall **87** and the front edge of the right wall **144**. The entry opening **93** has a slit shape that extends in the up-down direction. The entry opening **93** allows the ink ribbon **105** to enter the lower case **40** from the exposed area **92** therethrough.

[0089] The discharge opening **94** is formed in the front edge of the right wall **144**. The discharge opening **94** has a slit shape that extends in the up-down direction. The discharge opening **94** allows the printing tape **101** to be discharged from the lower case **40** therethrough. The discharge opening **94** is defined by a guide **145** at the front edge of the right wall **144**. The guide **145** has a solid cylindrical shape that is elongated in the up-down direction.

[0090] Next, a relationship between thicknesses of the main portion **84**A and the front wall **141** in the front-rear direction will be described with reference to FIG. **11**. First, a front surface area and a rear surface area will be defined.

[0091] The front wall **141** has a front surface **141**A, and a rear surface **141**B. The front surface **141**A is a surface of the front wall **141** that faces frontward. The rear surface **141**B is a surface of the front wall **141** that faces rearward. The front surface **141**A may include a plurality of surfaces having different positions in the front-rear direction, for example. Specifically, the front surface **141**A may include recessed portions or protruding portions. A front surface area is an area of the front surface **141**A within which all points in the front-rear direction fall in the same plane.

[0092] For example, only a single front surface area is present if the entire front surface **141**A is flush, but multiple front surface areas exists when the front surface **141**A has recessed portions or protruding portions. Specifically, if the front surface **141**A has both a recessed portion and a protruding portion, three surfaces serve as the front surface areas, i.e., a surface on a protruding end of the protruding portion, a surface in a bottom of the recessed portion, and a surface from which the protruding portion protrudes and in which an opening to the recessed portion is formed. The front surface **141**A includes a maximum front surface area. This maximum front surface area is the front surface area that occupies the largest amount of area within the front surface **141**A.

[0093] Next, the maximum front surface area will be described for one case. This case assumes that the front surface **141**A of the front wall **141** possesses one or more recessed portions and one or more protruding portions; a total area of the front surface area occupied by protruding ends of the one or more protruding portions is smaller than a total area of the front surface area occupied by a surface from which the protruding portions protrude and in which openings to the recessed portions are formed; and a total area of the front surface area occupied by bottom surfaces in the one or more recessed portions is smaller than the total area of the front surface area occupied by the surface from which the protruding portions protrude and in which the openings to the recessed portions are formed.

[0094] In this case, the front surface area occupied by the surface from which the protruding portions protrude and in which the openings to the recessed portions are formed is the front surface area occupying the greatest amount of area within the front surface **141**A and, hence, the maximum front surface area. FIG. **11** indicates a position of the maximum front surface area in the front-rear direction with a virtual line R1.

[0095] Similar to the front surface area, a rear surface area is an area of the rear surface **141**B within which all points in the front-rear direction fall in the same plane. The rear surface **141**B includes a maximum rear surface area. This maximum rear surface area is the rear surface area that occupies the largest amount of area within the rear surface **141**B. For example, the maximum rear surface area is an area of the rear surface **141**B excluding a path-defining part

98H (described later). FIG. 11 indicates a position of the maximum rear surface area in the front-rear direction with a virtual line R2.

[0096] The thickest portion of the main portion 84A in the front-rear direction will be referred to as "maximum thickness T1 of the main portion 84A". While not limited to any specific size, the maximum thickness T1 of the main portion 84A is 1.7 mm, for example. The maximum thickness T1 of the main portion 84A in the front-rear direction is greater than a distance D1 in the front-rear direction between the maximum front surface area and the maximum rear surface area. While not limited to any specific size, the distance D1 in the front-rear direction is 1.0 mm, for example.

[0097] The thinnest portion of the main portion 84A in the front-rear direction will be referred to as "minimum thickness T2 of the main portion 84A". While not limited to any specific size, the minimum thickness T2 of the main portion 84A is 1.7 mm, for example, which is the same as the maximum thickness T1 in the cassette 100. In other words, the main portion 84A in the cassette 100 of the present embodiment is a flat plate with uniform thickness. Thus, the minimum thickness T2 of the main portion 84A in the front-rear direction is greater than the distance D1 in the front-rear direction between the maximum front surface area and the maximum rear surface area.

[0098] Next, the paths of the ink ribbon 105 and the printing tape 101 will be described with reference to FIGS. 8 and 11. In addition to the path-defining parts 98A and 98B, the lower case 40 includes path-defining parts 98C, 98D, 98E, 98F, 98G, and 98H. The path-defining part 98C is provided on the front edge of the head right peripheral wall 87. The path-defining part 98C is a solid cylindrical member extending in the up-down direction and formed in a pin shape. During the conveyance of the ink ribbon 105, the ink ribbon 105 contacts the path-defining part 98C in the entirety of the width (i.e., in the entirety of the dimension in the up-down direction) of the ink ribbon 105 to have a contact portion Z5 (see FIG. 13) with the path-defining part 98C, and accordingly, the direction of the path of the ink ribbon 105 is changed. Specifically, the path of the ink ribbon 105 curves along an outer circumference of the path-defining part 98C. The contact portion Z5 between the ink ribbon 105 and the path-defining part 98C has a straight-line shape that extends in the up-down direction when viewed in the longitudinal direction of the ink ribbon 105.

[0099] The path-defining part 98D is provided on the rear edge of the head right peripheral wall 87. The path-defining part 98D is a solid cylindrical member extending in the up-down direction and formed in a pin shape. During the conveyance of the ink ribbon 105, the ink ribbon 105 contacts the path-defining part 98D in the entirety of the width (i.e., in the entirety of the dimension in the up-down direction) of the ink ribbon 105 to have a contact portion Z3 (see FIG. 13) with the path-defining part 98D, whereby the direction of the path of the ink ribbon 105 is changed. Specifically, the path of the ink ribbon 105 curves around an outer circumference of the path-defining part 98D. The contact portion Z3 between the ink ribbon 105 and the path-defining part 98D has a straight-line shape and extends in the up-down direction when viewed in the longitudinal direction of the ink ribbon 105.

[0100] The head right peripheral wall 87 includes a rib 97. The rib 97 is provided between the path-defining part 98C and the path-defining part 98D. That is, the rib 97 is provided on a portion of the head right peripheral wall 87 other than the front edge and the rear edge. The rib 97 is a protrusion that protrudes rightward from a right surface of the head right peripheral wall 87, and is elongated in the up-down direction. An upper end of the rib 97 is positioned further downward relative to an upper edge of the ink ribbon 105. A lower end of the rib 97 is positioned further upward relative to a lower edge of the ink ribbon 105. The center of the rib 97 in the up-down direction is approximately aligned with the center of the ink ribbon 105 in the up-down direction (i.e., the width direction). While not limited to any specific size, a length of the rib 97 in the up-down direction is approximately 7 mm, for example, while a width of the ink ribbon 105 in the cassette 100 is 12 mm.

[0101] A recess 97A is formed in the rib 97. The recess 97A is recessed leftward in the center of the rib 97. When the upper cover 80 is formed of resin, for example, the recess 97A allows gas generated during the injection molding process to escape.

[0102] The path-defining part 98E is provided in a left-rear portion of the lower case 40. The path-defining part 98E is a wall that protrudes downward from the intermediate wall 81. The path-defining part 98E extends frontward while extending leftward. The path-defining part 98E has a curved shape that expands rearward in the left-right center thereof. When the printing tape 101 is conveyed, the printing tape 101 contacts the path-defining part 98E so that a direction of the path of the printing tape 101 is changed. Specifically, the path of the printing tape 101 is curved along a rear surface of the path-defining part 98E.

[0103] The path-defining part 98F is provided in a left-front corner portion of the lower case 40. Specifically, the path-defining part 98F is disposed between the left edge of the head front peripheral wall 84 and the left wall 143. The path-defining part 98F is a roller extending in the up-down direction. The path-defining part 98F is rotatably supported by the lower case 40. During the conveyance of the printing tape 101, the printing tape 101 contacts the path-defining part 98F, whereby the direction of the path of the printing tape 101 is changed. Specifically, the path of the printing tape 101 is curved along an outer circumference of the path-defining part 98F.

[0104] The path-defining part 98G is provided at a position further rearward relative to the path-defining part 98D. The path-defining part 98G is a plate forming a wall structure that extends in both the up-down direction and the left-right direction. The path-defining part 98G protrudes leftward from the right wall 144. That is, the path-defining part 98G is provided on the opposite side of the path of the ink ribbon 105 from the path-defining part 98D when viewed in the up-down direction, and specifically on the right side of the path in the present embodiment. During the conveyance of the ink ribbon 105, the path-defining part 98G contacts the ink ribbon 105 in the entirety of the width (in the entirety of the dimension in the up-down direction) of the ink ribbon 105 to have a contact portion Z2 with the path-defining part 98G, thereby changing the direction of the path of the ink ribbon 105. Specifically, the path of the ink ribbon 105 is curved along a left edge of the path-defining part 98G. The contact portion Z2 between the ink ribbon 105 and the path-defining part 98G has a straight-line shape and extends in the up-down direction when viewed in the longitudinal direction of the ink ribbon 105. Note that the contact portion Z2 between the ink ribbon 105 and the path-defining part **98**G is positioned downstream of contact portion Z**3** between the ink ribbon **105** and the path-defining part **98**D in a conveying direction in which the ink ribbon **105** is conveyed.

[0105] The path-defining part **98**H is provided on the rear surface **141**B of the front wall **141**. The path-defining part **98**H is a solid cylindrical member extending in the up-down direction and is formed in a pin shape. When the printing tape **101** is conveyed, the path-defining part **98**H contacts the printing tape **101** to change the direction of the path of the printing tape **101**. Specifically, the path of the printing tape **101** is curved along an outer circumference of the path-defining part **98**H.

[0106] The path of the ink ribbon **105** is defined by the path-defining parts **98**A, **98**B, **98**C, **98**D, and **98**G. That is, the ink ribbon **105** extends frontward from the left side on an outer circumference of the ribbon roll **107** to the path-defining part **98**A while passing through a portion between the left wall **143** and the head left peripheral wall **86**. The ink ribbon **105** is curved rightward along the outer circumference of the path-defining part **98**A and passes into the arm portion **49**. Within the arm portion **49**, the ink ribbon **105** passes through a portion between the front wall **141** and the head front peripheral wall **84**, and specifically between the partitioning wall **88** and the head front peripheral wall **84**, and extends to the path-defining part **98**B.

[0107] Since the outer circumferences of the path-defining part **98**A and the path-defining part **98**B protrude frontward from the main portion **84**A, the ink ribbon **105** passes through the arm portion **49** at a position separated from and further frontward relative to the main portion **84**A and, hence, does not contact the main portion **84**A.

[0108] The ink ribbon **105** is guided into the exposed area **92** through the guide opening **91** while being curved diagonally rightward and rearward along the outer circumference of the path-defining part **98**B, and extends to the path-defining part **98**C. The ink ribbon **105** then re-enters the lower case **40** through the entry opening **93** while being curved rearward along the outer circumference of the path-defining part **98**C. From the path-defining part **98**C, the ink ribbon **105** extends rearward to pass through a portion between the head right peripheral wall **87** and the right wall **144**, and extends to the path-defining part **98**D.

[0109] The ink ribbon **105** contacts the rib **97** at a position between the path-defining part **98**C and the path-defining part **98**D to have a contact portion **Z1** with the rib **97**. At this time, the rib **97** contacts the widthwise center portion of the ink ribbon **105** (the center portion of the ink ribbon **105** in the width direction, i.e., in the up-down direction) at the contact portion **Z1** but does not contact either widthwise edge (either edge in the widthwise direction), as illustrated in FIG. **12**. Therefore, the rib **97** causes the ink ribbon **105** to be curved along the width of the ink ribbon **105** with both the widthwise edges (upper and lower edges) of the ink ribbon **105** curled leftward, i.e., with the widthwise center portion of the ink ribbon **105** protruding rightward. This shape of the ink ribbon **105** suppresses skewing of the ink ribbon **105**. Note that skewing of the ink ribbon **105** is a phenomenon in which the ink ribbon **105** progresses along a direction not orthogonal to the axis AX**1** (i.e., the up-down direction). Directions that are not orthogonal to the axis AX**1** include direction sloping downward as extending rightward and direction sloping downward as extending rearward.

[0110] In the present embodiment, the path-defining part **98**C is present between the exposed area **92** and the rib **97**. Therefore, even when the ink ribbon **105** formed in a curve along the width direction by the rib **97**, the ink ribbon **105** is returned to its uncurved state by the path-defining part **98**C. Accordingly, this arrangement suppresses curling of the ink ribbon **105** in the width direction produced by the rib **97** from adversely affecting printing by the thermal head **4** in the exposed area **92**.

[0111] As illustrated in FIGS. **8** and **11**, the ink ribbon **105** is curved rearward and slightly leftward along the outer circumference of the path-defining part **98**D and extends to the path-defining part **98**G. The ink ribbon **105** is then curved rearward and slightly rightward along the left edge of the path-defining part **98**G and crosses over to the right edge on the circumference of the take-up spool **109**. In other words, the ink ribbon **105** passes from the ribbon roll **107** sequentially over the path-defining part **98**A, the path-defining part **98**B, the path-defining part **98**C, the path-defining part **98**D, and the path-defining part **98**G and extends to the take-up spool **109**.

[0112] The take-up spool **109** rotates in a clockwise direction in a bottom view to take up the ink ribbon **105**. Hence, after passing through a portion between the head right peripheral wall **87** and the right wall **144**, the ink ribbon **105** is maintained between the center of the take-up spool **109** and the right wall **144** in the left-right direction while being taken up by the take-up spool **109**.

[0113] Note that, in a case where the take-up spool **109** is configured to take up the ink ribbon **105** by rotating in a counterclockwise direction in a bottom view, the ink ribbon **105** would cross over to the left edge on the outer circumference of the take-up spool **109** after passing through a portion between the head right peripheral wall **87** and the right wall **144**. For this reason, after passing through a portion between the head right peripheral wall **87** and the right wall **144**, the ink ribbon **105** is taken up by the take-up spool **109** by crossing from right to left through a portion between the center of the take-up spool **109** and the path-defining part **98**D.

[0114] In other words, the lower case **40** defines the path along which the ink ribbon **105** is conveyed in the conveying direction when the take-up spool **109** takes up the ink ribbon **105** from the ribbon roll **107**. The path of the ink ribbon **105** includes a first part, a second part, the exposed area **92**, a third part, and a fourth part from an upstream side in the conveying direction. The first part passes between the left wall **143** and the head left peripheral wall **86**. The second part passes between the front wall **141** and the head front peripheral wall **84**. The third part passes between the head right peripheral wall **87** and the right wall **144**. The fourth part passes between the right wall **144** and the center of the take-up spool **109**.

[0115] The path for the printing tape **101** is defined by the path-defining parts **98**E, **98**F, and **98**H. That is, the printing tape **101** extends diagonally downward and leftward from a rear edge on an outer circumference of the tape roll **103**, is inserted sequentially through the tape guide slit (not illustrated) and the tape insertion opening **44**, and extends to the path-defining part **98**E. The printing tape **101** is then curved frontward along an outer circumference of the path-defining part **98**E, passes through a portion between the left wall **143** and the head left peripheral wall **86**, and extends to the path-defining part **98**F. The printing tape **101** is curved rightward along an outer circumference of the path-defining part **98**F and passes into the arm portion **49**.

[0116] Within the arm portion **49**, the printing tape **101** extends from left to right while curving slightly frontward along an outer circumference of the path-defining part **98**H. While in the arm portion **49**, the printing tape **101** passes through a portion between the front wall **141** and the head front peripheral wall **84**, and more specifically between the front wall **141** and the partitioning wall **88**, and extends to the guide opening **91**. The printing tape **101** is guided into the exposed area **92** through the guide opening **91**.

[0117] In the exposed area **92**, the ink ribbon **105** is superimposed with the printing tape **101** from rearward. The printing tape **101** then extends from the exposed area **92** to the discharge opening **94** and is discharged from the case **20** through the discharge opening **94**. In other words, the printing tape **101** passes sequentially from the tape roll **103** over the path-defining part **98**E, the path-defining part **98**F, and the path-defining part **98**H and is discharged through the discharge opening **94**.

[0118] In other words, the lower case **40** further defines the path along which the printing tape **101** is conveyed in a conveying direction of the printing tape **101**. The path of the printing tape **101** includes the tape insertion opening **44**, a first part, a second part, and the exposed area **92** from an upstream side in the conveying direction of the printing tape **101**. The first part of the path for the printing tape **101** passes between the left wall **143** and the head left peripheral wall **86**. The second part of the path for the printing tape **101** passes between the front wall **141** and the head front peripheral wall **84**.

[0119] Here, positional relationships among the take-up spool **109**, the path-defining parts **98**D and **98**G, and the rib **97** will be described in greater detail with reference to FIG. **13**. A virtual line S**1** in FIG. **13** is a straight line that passes through both the contact portion Z**1** between the rib **97** and the ink ribbon **105** and the contact portion Z**2** between the path-defining part **98**G and the ink ribbon **105** when viewed in the up-down direction. Among a pair of straight lines tangential to the outer circumferential surface of the take-up spool **109** and passing through the contact portion Z**3** between the path-defining part **98**D and the ink ribbon **105**, a virtual line S**2** denotes the straight line passing through the right side of the take-up spool **109**. That is, the virtual line S**2** denotes the straight line passing through the contact portion Z**3** between the path-defining par **98**D and the ink ribbon **105** and tangential to the outer circumferential surface of the take-up spool **109** at a position rightward of the center of the take-up spool **109**. The virtual line S**2** passes through a point Z**4** on the outer circumferential surface of the take-up spool **109**. That is, the point Z**4** is the point of tangency between the virtual line S**2** and the outer circumferential surface of the take-up spool **109** and is positioned rightward relative to the center of the take-up spool **109**.

[0120] When viewed in the up-down direction, the virtual line S**1** passes through a point positioned to the left of the contact portion Z**3** between the path-defining part **98**D and the ink ribbon **105**. In other words, the contact portion Z**3** between the path-defining part **98**D and the ink ribbon **105** is positioned on the right side of the virtual line S**1**. When viewed in the up-down direction, the virtual line S**2** passes through a point positioned to the right of the contact portion Z**2** between the path-defining part **98**G and the ink ribbon **105**. In other words, the contact portion Z**2** between the path-defining part **98**G and the ink ribbon **105** is positioned to the left of the virtual line S**2**.

[0121] Next, a method of fixing the printing tape **101** to the tape spool **102** will be described with reference to FIGS. **14**A and **14**B. The printing tape **101** is fixed to the tape spool **102** via an anchoring tape **104**. Specifically, the anchoring tape **104** has a proximal end **104**A and a distal end **104**B. The proximal end **104**A is fixed to an outer circumferential surface of the tape spool **102**. The distal end **104**B of the anchoring tape **104** is attached to a proximal end **101**A of the printing tape **101** via an adhesive tape **108**. The anchoring tape **104** and the adhesive tape **108** do not constitute the printing medium.

[0122] The anchoring tape **104** and the adhesive tape **108** have colors that allow the user to recognize the presence of the anchoring tape **104** and the adhesive tape **108** and to recognize that the anchoring tape **104** and the adhesive tape **108** are different from the printing tape **101**. For example, the anchoring tape **104** has a pattern such as zebra stripes or other colors that are different from a color of the printing tape **101**. The adhesive tape **108** also has a different color from the color of the printing tape **101** and is not colorless and transparent. For example, the adhesive tape **108** is red. The color and pattern of the adhesive tape **108** may be the same as or different from the color and pattern of the anchoring tape **104**.

[0123] A virtual line Q in FIGS. **14**A and **14**B indicates a position of the guide opening **91** relative to the tape spool **102**. A length L**1** of the anchoring tape **104** in the longitudinal direction is approximately equivalent to a distance L**2** from the tape spool **102** to the guide opening **91** (the virtual line Q) along the conveying path of the printing tape **101**. When all of the printing tape **101** has been drawn off the tape spool **102**, the distal end **104**B of the anchoring tape **104** is exposed outside through the guide opening **91**, as illustrated in FIG. **14**A. That is, the distal end **104**B of the anchoring tape **104** is positioned further rightward relative to the virtual line Q. At this time, the user can recognize that the cassette **100** has run out of the printing tape **101**.

[0124] As illustrated in FIG. **14**B, the distal end **104**B of the anchoring tape **104** may not reach the guide opening **91** while the anchoring tape **104** has not been completely drawn off the tape spool **102**, for example. In other words, the distal end **104**B of the anchoring tape **104** may be positioned further leftward relative to the virtual line Q at this time. In such cases, if the adhesive tape **108** were colorless and transparent, the user would have difficulty seeing either of the anchoring tape **104** and the adhesive tape **108**. Therefore, it would be difficult for the user to recognize that the cassette **100** has run out of printing tape **101**.

[0125] With the cassette **100** according to the present embodiment, the user can recognize that the cassette **100** has run out of the printing tape **101** once the adhesive tape **108** has reached the guide opening **91**, even if the distal end **104**B of the anchoring tape **104** has not arrived at the guide opening **91**. The length L**1** of the anchoring tape **104** may be difficult to modify due to design, manufacturing, or other concerns. Thus, by utilizing the adhesive tape **108**, the cassette **100** of the first embodiment helps let the user know when the cassette **100** has run out of the printing tape **101** without having to adjust the length L**1** of the anchoring tape **104**.

[0126] Next, printing operations on the printer **1** will be described with reference to FIGS. **1** through **3**. While the cover **3** illustrated in FIG. **1** is removed from the housing **2**, as illustrated in FIGS. **1** and **2**, the user attaches the cassette **100** to the attachment portion **11** from above so that the rotational shaft **6** is inserted into the lower support portion **41** (see FIG. **4**) and the thermal head **4** is inserted into the head opening **99**. The user then attaches the cover **3** to the housing **2**, whereby a cam (not illustrated) causes the right end of the platen holder **15** to be pivotally moved rearward about the pivot shaft **16** from a position in which the platen holder **15** is urged frontward by the spring (not illustrated). At this time, the platen roller **18** illustrated in FIG. **1** presses the ink ribbon **105** and the printing tape **101** in their overlapped state against the thermal head **4** from frontward. Additionally, the platen gear **19** is brought into meshing engagement with the output gear **21**.

[0127] The user inputs a print instruction into the printer **1** through an operation unit (not illustrated). The printer **1** drives the conveying motor to rotate the rotational shaft **6**, whereby the rotational shaft **6** rotates the take-up spool **109** and the input gear **23**. The rotation of the take-up spool **109** conveys the ink ribbon **105** from the ribbon roll **107** toward the take-up spool **109**. The rotation of the input gear **23** rotates the output gear **21** via the coupling gear **22**. The rotation of the output gear **21** rotates both the platen gear **19** and the platen roller **18**.

[0128] By rotating in the clockwise direction in a plan view, the platen roller **18** draws the printing tape **101** off the tape roll **103** and conveys the printing tape **101** rightward. Hence, the left side of the printer **1** is upstream in the conveying direction of the printing tape **101** while the right side of the printer **1** is downstream in the conveying direction of the printing tape **101**. The thermal head **4** selectively heats a plurality of heating elements, resulting in ink in the ink ribbon **105** being thermally transferred onto the printing tape **101** at positions corresponding to the heated heating elements.

[0129] The printer **1** performs printing on the printing tape **101** in the exposed area **92** by repeatedly performing the conveyance of the printing tape **101** with the platen roller **18** and selectively heating of the plurality of heating elements with the thermal head **4**. The printed printing tape **101** is discharged from the lower case **40** through the discharge opening **94** and then discharged from the housing **2** through the discharge opening **12**.

[0130] In the first embodiment described above, the maximum thickness T**1** of the main portion **84**A in the front-rear direction is greater than the distance D**1** between the maximum front surface area and the maximum rear surface area in the front-rear direction. Accordingly, the head front peripheral wall **84** can be made more rigid than a configuration in which the maximum thickness T**1** of the main portion **84**A in the front-rear direction is smaller than or equal to the distance D**1** between the maximum front surface area and the maximum rear surface area in the front-rear direction. This configuration reduces the likelihood that the lower edge of the head front peripheral wall **84** is bent rearward relative to the upper edge of the head front peripheral wall **84**. Therefore, the cassette **100** can help suppress skewing of the ink ribbon **105**.

[0131] In the first embodiment, the minimum thickness T**2** of the main portion **84**A in the front-rear direction is greater than the distance D**1** between the maximum front surface area and maximum rear surface area in the front-rear direction, thereby further enhancing the rigidity of the main portion **84**A. Accordingly, the cassette **100** can better restrain skewing of the ink ribbon **105**.

[0132] In the first embodiment, the tape insertion opening **44** is formed in the intermediate wall **81**. With this arrangement, the user can more easily handle the printing tape **101** than a configuration in which the printing tape **101** is handled separately from the lower case **40**. Accordingly, the cassette **100** can help facilitate handling of the printing tape **101**.

[0133] In the first embodiment, the cassette **100** includes the upper case **30**. The user can handle the printing tape **101** without touching the same. Accordingly, the cassette **100** can help facilitate handling of the printing tape **101**.

[0134] In the first embodiment, the rib **46**A protrudes rearward from the opening edge **46**. Therefore, when a force in a direction for bending the lower edge of the head front peripheral wall **84** rearward relative to the upper edge of the head front peripheral wall **84** is applied to the lower wall **71**, the rib **46**A contacts the lower end **88**A of the partitioning wall **88**. At this time, the force in the direction for bending the lower edge of the head front peripheral wall **84** rearward relative to the upper edge of the head front peripheral wall **84** is distributed from the lower wall **71** to the partitioning wall **88** via the rib **46**A. This action restrains the lower edge of the head front peripheral wall **84** from bending rearward relative to the upper edge of the head front peripheral wall **84**. Accordingly, the cassette **100** helps suppress skewing of the ink ribbon **105**.

[0135] In the first embodiment, the rib **97** protrudes rightward from a portion of the head right peripheral wall **87** excluding the front edge and the rear edge. The rib **97** contacts the widthwise center of the ink ribbon **105**, thereby forming a curve in the ink ribbon **105** along the width direction. By employing the rib **97** to curl the ink ribbon **105** along the width direction in this way, any inclination of the ink ribbon **105** to deviate in the width direction applies a force to the ink ribbon **105** that suppresses such deviation. Accordingly, the cassette **100** can help suppress skewing of the ink ribbon **105**.

[0136] In the first embodiment, the path-defining part **98**D is provided on the rear edge of the head right peripheral wall **87**. The path-defining part **98**D contacts the ink ribbon **105** along the entire width thereof. This contact portion between the part-defining part **98**D and the ink ribbon **105** is in a form of a straight line extending in the up-down direction when viewed in the longitudinal direction of the ink ribbon **105**. Hence, after the ink ribbon **105** has been formed in a curve along the width direction by the rib **97**, the path-defining part **98**D returns the ink ribbon **105** to an uncurved state in the width direction. As a result, the ink ribbon **105** taken up on the take-up spool **109** is not curved in the width direction. Thus, the cassette **100** can help restrain the ink ribbon **105** from being taken up by the take-up spool **109** while still curled along the width direction, thereby avoiding defects on winding of the ink ribbon **105** on the take-up spool **109**.

[0137] In the first embodiment, the virtual line **S1** passes through a point positioned to the left of the contact portion **Z3** between the path-defining part **98**D and the ink ribbon **105** when viewed in the up-down direction. Therefore, the path-defining part **98**G presses the ink ribbon **105** leftward against the path-defining part **98**D. This pressure by the path-defining part **98**G causes the rib **97** to further make the ink ribbon **105** into a curved shape along the width direction. Hence, the cassette **100** further helps to suppress skewing of the ink ribbon **105**.

[0138] In the first embodiment, the virtual line S**2** passes through a point positioned to the right of the contact portion Z**2** between the path-defining part **98**G and the ink ribbon **105** when viewed in the up-down direction. Therefore, the path-defining part **98**G presses the ink ribbon **105** leftward against the path-defining part **98**D, regardless of the size of the outer diameter of a roll configured of the ink ribbon **105** taken up on the take-up spool **109**. As a result, the rib **97** causes the ink ribbon **105** to be curved even more along the width direction. Accordingly, the cassette **100** can further suppress skewing of the ink ribbon **105**.

[0139] In the first embodiment, the length of the head right peripheral wall **87** in the front-rear direction is smaller than the length of the head rear peripheral wall **85** in the left-right direction. This configuration shortens a distance from the rib **97** to the path-defining part **98**D so that the rib **97** causes the ink ribbon **105** to be curved even more along the width direction. Accordingly, the cassette **100** can further help suppress skewing of the ink ribbon **105**.

[0140] In the first embodiment, in the radial direction of the take-up spool **109**, the first gap G**1** is formed between the lower end of the take-up spool **109** and the lower support portion **41**, and the second gap G**2** is formed between the upper end of the take-up spool **109** and the upper support portion **43**. FIG. **15**A illustrates a case in which, unlike in the first embodiment, the first gap G**1** and the second gap G**2** illustrated in FIG. **6** have sizes the same as each other. FIGS. **15**B and **15**C illustrate a case in which the first gap G**1** of FIG. **6** is greater than the second gap G**2** of FIG. **6**, as in the first embodiment.

[0141] In the example of FIG. **15**A, the take-up spool **109** is moved rightward when a tensile force F**1** acts on the ink ribbon **105** in a direction that slopes upward toward rightward. In this case, the lower end of the take-up spool **109** contacts the lower support portion **41** on the right side, and the upper end of the take-up spool **109** contacts the upper support portion **43** on the left side. The contact between the lower end of the take-up spool **109** and the lower support portion **41** and the contact between the upper end of the take-up spool **109** and the upper support portion **43** are nearly simultaneous. Accordingly, the take-up spool **109** rotates while the axis AX**3** thereof remains aligned in the up-down direction.

[0142] Therefore, in a case where the first gap G**1** and the second gap G**2** have sizes the same as each other, the ink ribbon **105** is wound about the take-up spool **109** while being gradually shift upward in the axial direction of the take-up spool **109** when the take-up spool **109** takes up the ink ribbon **105** with the tensile force F**1** acting on the ink ribbon **105**. This action may result in defects in winding of the ink ribbon **105** by the take-up spool **109**, such as the ink ribbon **105** contacting the guide disc **109**B.

[0143] In the first embodiment, the size of the first gap G**1** differs from the size of the second gap G**2**. Hence, when the take-up spool **109** is moved rightward due to the tensile force F**1** acting on the ink ribbon **105**, as illustrated in FIG. **15**B, the upper end of the take-up spool **109** contacts the upper support portion **43** on the left side before the lower end of the take-up spool **109** contacts the lower support portion **41**. In other words, the contact between the lower end of the take-up spool **109** and the lower support portion **41** is not simultaneous with the contact between the upper end of the take-up spool **109** and the upper support portion **43**.

[0144] Hence, at the point in time that the upper end of the take-up spool **109** contacts the upper support portion **43**, a gap exists between the lower end of the take-up spool **109** and the lower support portion **41**. Accordingly, the lower end of the take-up spool **109** is pivotally moved rightward about the contact portion between the upper end of the take-up spool **109** and the upper support portion **43** on the left side so that the axis AX**3** becomes orthogonal to the direction of the tensile force F**1**, as illustrated in FIG. **15**C. As a result, the take-up spool **109** is placed in a state in which the upper end of the take-up spool **109** contacts the upper support portion **43** on the right side and the lower end of the take-up spool **109** contacts the lower support portion **41** on the right side, for example.

[0145] When the take-up spool **109** takes up the ink ribbon **105** in this state, the ink ribbon **105** is restrained from shifting in the axial direction of the take-up spool **109** when being taken up thereon. Since the size of the first gap G**1** differs from the size of the second gap G**2** in the first embodiment, the axis AX**3** of the take-up spool **109** tilts along the width direction of the ink ribbon **105** even when the ink ribbon **105** becomes skewed. Accordingly, the cassette **100** can help suppress defective winding of the ink ribbon **105** by the take-up spool **109**.

[0146] In the first embodiment, the first gap G**1** is greater than the second gap G**2**. This facilitates an operator in assembling the lower cover **70** with the upper cover **80** for the following reason. The operator assembles the lower cover **70** on the upper cover **80** after mounting the take-up spool **109** in the upper cover **80**. If the second gap G**2** were greater than the first gap G**1**, the operator would need to assemble the lower cover **70** on the upper cover **80** while the take-up spool **109** supported by the upper support portion **43** is very wobbly.

[0147] Since the second gap G**2** is smaller than the first gap G**1** in the first embodiment, the operator can assemble the lower cover **70** on the upper cover **80** with less wobble in the take-up spool **109** supported by the upper support portion **43**. This arrangement reduces the likelihood of the lower cover **70** colliding with the lower end of the take-up spool **109** from below. Accordingly, the cassette **100** helps to facilitate assembly of the lower cover **70** on the upper cover **80**.

[0148] Further, since the first gap G**1** is greater than the second gap G**2**, the user can easily attach the cassette **100** to the attachment portion **11** for the following reason. The user attaches the cassette **100** to the attachment portion **11** so that the rotational shaft **6** is fitted into the take-up spool **109**. Hence, the rotational shaft **6** first passes through the lower end of the take-up spool **109** and subsequently passes through the upper end of the take-up spool **109**. If the first gap G**1** were formed smaller than the second gap G**2**, the user would need to attach the cassette **100** to the attachment portion **11** with only a small amount of play in the take-up spool **109** supported by the lower support portion **41**.

[0149] Since the first gap G**1** is greater than the second gap G**2** in the first embodiment, the user can attach the cassette **100** to the attachment portion **11** with a greater amount of play in the take-up spool **109** supported by the lower support portion **41**. This arrangement reduces the likelihood that the lower end of the take-up spool **109** will collide with the rotational shaft **6** from above when the rotational shaft **6** is guided into the take-up spool **109**. Accordingly, the cassette **100** helps facilitate the user in attaching the cassette **100** to the attachment portion **11**.

[0150] In a case where the absolute value of the difference between the first gap G**1** and the second gap G**2** is greater than the thickness T**3** of the lower support portion **41** in the up-down direction, the take-up spool **109** may come out of the lower support portion **41** when the take-up spool **109** is tilted. In the first embodiment, the absolute value of the difference between the first gap G**1** and the second gap G**2** is smaller than the thickness T**3** of the lower support portion **41** in the up-down direction. Therefore, the cassette **100** can help suppress potential for the take-up spool **109** falling out of the lower support portion **41**.

[0151] In a case where the absolute value of the difference between the first gap G**1** and the second gap G**2** is greater than the thickness T**4** of the upper support portion **43** in the up-down direction, the take-up spool **109** may come out of the upper support portion **43** when the take-up spool **109** is tilted. In the first embodiment, the absolute value of the difference between the first gap G**1** and the second gap G**2** is smaller than the thickness T**4** of the upper support portion **43** in the up-down direction. Therefore, the cassette **100** helps reduce the likelihood of the take-up spool **109** coming out of the upper support portion **43**.

[0152] In the first embodiment, the lower support portion **41** is arranged outside the outer circumference of the take-up spool **109** in the radial direction of the take-up spool **109**. Therefore, the lower wall **71** is lighter than a configuration in which the lower support portion **41** is arranged inside the inner circumference of the take-up spool **109** in the radial direction, because a hole defined by the lower support portion **41** is larger. In the first embodiment, the upper support portion **43** is arranged inside the inner circumference of the take-up spool **109** in the radial direction of the take-up spool **109**. This configuration of the cassette **100** ensures that the thickness T**4** of the upper support portion **43** in the up-down direction is sufficiently thick.

[0153] Accordingly, the cassette **100** helps reduce the likelihood of the take-up spool **109** coming out of the upper support portion **43**. In other words, the configuration of the cassette **100** makes the case **20** lighter while helping to reduce the likelihood of the take-up spool **109** coming out of the upper support portion **43**. Further, since the upper support portion **43** is arranged radially inside the inner circumference of the take-up spool **109**, the cassette **100** ensures that there is sufficient space for mounting the clutch spring **119** around the upper end of the take-up spool **109**.

[0154] The first embodiment may be modified in various ways. Here, a cassette **100**A according to a modification of the first embodiment will be described with reference to FIG. **16**. In this modification, the cassette **100**A includes a head front peripheral wall **184** in place of the head front peripheral wall **84**. The head front peripheral wall **184** includes a protruding portion **84**B in addition to the main portion **84**A. The protruding portion **84**B protrudes rearward from a rear surface of the main portion **84**A, and extends in both the up-down direction and the left-right direction. The protruding portion **84**B has a plate shape.

[0155] A length of the protruding portion **84**B in the left-right direction is shorter than a length of the main portion **84**A in the left-right direction. A length of the protruding portion **84**B in the up-down direction is either the same as or smaller than a length of the main portion **84**A in the up-down direction. A length of the protruding portion **84**B in the front-rear direction is smaller than the length of the protruding portion **84**B in the left-right direction and smaller than the length of the protruding portion **84**B in the up-down direction.

[0156] In the cassette **100**A of this modification, the maximum thickness T**11** of the main portion **84**A in the front-rear direction is defined by a position in which the protruding portion **84**B is present. In other words, the maximum thickness T**11** of the main portion **84**A in the front-rear direction is equal to a distance in the front-rear direction between a rear edge of the protruding portion **84**B and a front surface of the main portion **84**A. The maximum thickness T**11** of the main portion **84**A in the front-rear direction is greater than the distance DI in the front-rear direction between the maximum front surface area and the maximum rear surface area.

[0157] The minimum thickness T**12** of the main portion **84**A in the front-rear direction is defined by positions in which the protruding portion **84**B is not present. In other words, the minimum thickness T**12** of the main portion **84**A in the front-rear direction is equal to a distance in the front-rear direction between the rear surface of the main portion **84**A in a portion where the protruding portion **84**B is not present and the front surface of the main portion **84**A. In the cassette **100**A of this modification, the minimum thickness T**12** of the main portion **84**A in the front-rear direction is either the same as or smaller than the distance D**1** in the front-rear direction between the maximum front surface area and the maximum rear surface area.

[0158] With the cassette **100**A having the above configuration, the weight of the main portion **84**A is lighter than a configuration in which the minimum thickness T**12** of the main portion **84**A in the front-rear direction is greater than the distance D**1** in the front-rear direction between the maximum front surface area and the maximum rear surface area. On the other hand, the head front peripheral wall **84** has higher rigidity since the maximum thickness T**11** of the head front peripheral wall **84** in the front-rear direction is greater than the distance D**1** in the front-rear direction between the maximum front surface area and the maximum rear surface area. Hence, the cassette **100**A having this configuration contributes to make the lower case **40** lighter while suppressing skewing of the ink ribbon **105**.

[0159] Next, a cassette **100**B according to another modification of the first embodiment will be described with reference to FIG. **17**. In this modification, the cassette **100**B includes a head front peripheral wall **284** in place of the head front peripheral wall **84** or the head front peripheral wall **184**. The head front peripheral wall **284** includes a plurality of ribs **84**C. The ribs **84**C protrude rearward from the rear surface of the main portion **84**A and extend in the up-down direction. The ribs **84**C are arranged in the left-right direction.

[0160] In the cassette **100**B according to this modification, the maximum thickness T**21** of the main portion **84**A in the front-rear direction is defined at positions of the ribs **84**C. In other words, the maximum thickness T**21** of the main portion **84**A in the front-rear direction is equal to a distance in the front-rear direction between rear edges of the ribs **84**C and the front surface of the main portion **84**A. The maximum thickness T**21** of the main portion **84**A in the front-rear direction is greater than the distance D1 in the front-rear direction between the maximum front surface area and the maximum rear surface area.

[0161] The minimum thickness T22 of the main portion **84**A in the front-rear direction is defined at positions of recesses formed between adjacent ribs **84**C. In other words, the minimum thickness T22 of the main portion **84**A in the front-rear direction is equal to a distance in the front-rear direction between the rear surface of the main portion **84**A between the ribs **84**C (i.e., bottom surfaces of recessed portions formed by the ribs **84**C) and the front surface of the main portion **84**A. With the cassette **100**B having this configuration, the minimum thickness T22 of the main portion **84**A in the front-rear direction is either the same as or smaller than the distance D1 in the front-rear direction between the maximum front surface area and the maximum rear surface area.

[0162] The head front peripheral wall **284** in this cassette **100**B is even lighter than the head front peripheral wall **184** in the cassette **100**A. Accordingly, the cassette **100**B can make the lower case **40** even lighter while suppressing skewing of the ink ribbon **105**.

[0163] Next, other modifications to the first embodiment will be described. The rib **46**A may protrude frontward from a rear edge of the opening edge **46**, or may be omitted from the lower case **40**. The upper case **30** may be omitted from the cassette **100**. In this case, the tape roll **103** may be supported in the lower case **40**. The tape roll **103** may be omitted from the cassette **100**.

[0164] The tape insertion opening **44** may not be formed in the intermediate wall **81**. In this case, the tape roll **103** may be accommodated in the lower case **40**, for example. The upper bridging portion **81**A may have a columnar shape that is elongated in the front-rear direction.

[0165] The path-defining part **98**D need not be provided on the rear edge of the head right peripheral wall **87**, but may be provided between the rear edge of the head right peripheral wall **87** and the path-defining part **98**G in the front-rear direction, for example. The virtual line S1 may pass through a point positioned on the right side of the contact portion Z3 between the path-defining part **98**D and the ink ribbon **105** when viewed in the up-down direction. The virtual line S2 may pass through a point positioned on the left side of the contact portion Z2 between the path-defining part **98**G and the ink ribbon **105** when viewed in the up-down direction.

[0166] One or both of the path-defining part **98**D and the path-defining part **98**G may be omitted, for example. In a case where both the path-defining part **98**D and the path-defining part **98**G are omitted, the ink ribbon **105** is taken up directly on the take-up spool **109** after passing over the rib **97**. The length of the head right peripheral wall **87** in the front-rear direction may be the same as or greater than the length of the head rear peripheral wall **85** in the left-right direction.

[0167] The first gap G1 may have a size the same as that of the second gap G2. Alternatively, the first gap G1 may be smaller than the second gap G2. The absolute value of the difference between the first gap G1 and the second gap G2 may be the same as or greater than the thickness T3 of the lower support portion **41** in the up-down direction. The absolute value of the difference between the first gap G1 and the second gap G2 may be the same as or greater than the thickness T4 of the upper support portion **43** in the up-down direction. The outer diameter R11 at the lower end of the take-up spool **109** may be the same as or smaller than the inner diameter R12 at the upper end of the take-up spool **109**.

[0168] The lower support portion **41** may be provided inside the inner circumference of the take-up spool **109** in the radial direction of the take-up spool **109**. The upper support portion **43** may be provided outside the outer circumference of the take-up spool **109** in the radial direction of the take-up spool **109**. Both the lower support portion **41** and the upper support portion **43** may be arranged radially outside the outer circumference of the take-up spool **109**, or radially inside the inner circumference of the take-up spool **109**. In a case where the lower support portion **41** is arranged inside the take-up spool **109**, the lower support portion **41** may protrude upward from the lower wall **71**, for example.

[0169] The take-up spool **109** may take up the ink ribbon **105** by rotating in a counterclockwise direction when viewed from below. In this case, the path-defining part **98**G may be arranged such that, of the pair of tangents to the outer circumference surface of the take-up spool **109** that pass through the contact portion Z3 between the path-defining part **98**D and the ink ribbon **105**, the tangent passing along the left of the take-up spool **109** passes on the right side of the contact portion Z2 between the path-defining part **98**G and the ink ribbon **105**.

Second Embodiment

[0170] Next, a cassette **200** according to a second embodiment of the present disclosure will be described with reference to FIGS. **18** and **19**. Parts and components in the cassette **200** having the same shape or function as those in the cassette **100** will be designated with the same reference numerals, or the reference numerals will be omitted for such parts and components to omit or simplify their description. In the cassette **200** according to the second embodiment, the maximum thickness of the main portion **84**A in the front-rear direction may be greater than, smaller than, or the same as a distance in the front-rear direction between the maximum front surface area and the maximum rear surface area.

[0171] The cassette **200** is for use in a printer **1**, and is used in the same manner as the cassette **100**. The cassette **200** differs from the cassette **100** in that the cassette **200** includes a lower wall **271** in place of the lower wall **71**. All configurations of the cassette **200** other than the lower wall **271** are the same as those of the cassette **100**.

[0172] The lower wall **271** has a shape different from the lower wall **71**. Specifically, the lower wall **271** includes a lower bridging portion **272** in addition to the structure of the lower wall **71** in the first embodiment. The lower bridging portion **272** is a part of the lower wall **271**. Specifically, the lower bridging portion **272** is a portion that spans across and covers from below a part of a space defined by the head front peripheral wall **84**, the head rear peripheral wall **85**, the head left peripheral wall **86**, and the head right peripheral wall **87**, i.e., a part of the head opening **99**. The lower bridging portion **272** is a wall configuration that extends in both the front-rear direction and the left-right direction.

[0173] Here, a centerline C1 will be defined. The centerline C1 is a virtual line extending in the front-rear direction and bisecting the head front peripheral wall **84** in the left-right direction. Next, a front trisector V1 and a rear trisector V2 will be defined. The front trisector V1 and the rear trisector V2 extend in the left-right direction and trisect the head left peripheral wall **86** in the front-rear direction. The front trisector V**1** and the rear trisector V**2** are arranged from front to rear in this order.

[0174] In the following description, a point P**1** will refer to a position between the left edge and the right edge of the head front peripheral wall **84** in the left-right direction. In the second embodiment, the point P**1** is positioned further leftward relative to the centerline C**1**. Additionally, a point P**2** will refer to a position between the left edge and the right edge of the head rear peripheral wall **85** in the left-right direction or between the front edge and the rear edge of the head left peripheral wall **86** in the front-rear direction. In the second embodiment, point P**2** is positioned further rearward relative to the front trisector V**1**.

[0175] A right-rear edge **272**A of the lower bridging portion **272** is connected to a front edge **711** of the second notch **71**B at the point P**1**. The right-rear edge **272**A extends from the point P**1** to the point P**2**. In the second embodiment, the right-rear edge **272**A extends diagonally rearward and leftward from the point P**1**. The right-rear edge **272**A is connected to one of a rear edge **712** and a left edge **713** of the second notch **71**B at the point P**2**. In the second embodiment, the right-rear edge **272**A is connected to the left edge **713** of the second notch **71**B at the point P**2**.

[0176] As described above, the lower case **40** of the second embodiment includes the lower bridging portion **272**. With this configuration, if a force directed rearward acts on the lower edge of the head front peripheral wall **84**, the lower bridging portion **272** receives this applied force. This configuration suppresses the lower edge of the head front peripheral wall **84** from bending rearward relative to the upper edge of the head front peripheral wall **84**. Accordingly, the cassette **200** can help to suppress skewing of the ink ribbon **105**.

[0177] In the second embodiment, the point P**2** is located further rearward relative to the front trisector V**1**. With this arrangement, the lower bridging portion **272** can receive a greater amount of the force directed rearward and acting on the head front peripheral wall **84** than a configuration in which the point P**2** is positioned further frontward relative to the front trisector V**1**. Hence, this configuration can better suppress the lower edge of the head front peripheral wall **84** from bending rearward relative to the upper edge of the head front peripheral wall **84**, whereby the cassette **200** can further help to suppress skewing of the ink ribbon **105**.

[0178] In the second embodiment, the point P**1** is positioned further leftward relative to the centerline C**1**. This configuration allocates more space for the head opening **99** in the left-right direction than a configuration in which the point P**1** is located further rightward relative to the centerline C**1**. Accordingly, the cassette **200** can ensure sufficient space for the head opening **99** while suppressing skewing of the ink ribbon **105**.

[0179] As in the first embodiment described above, the tape insertion opening **44** is formed in the intermediate wall **81** of the second embodiment. Accordingly, the cassette **200** can help facilitate handling of the printing tape **101**, as in the first embodiment. As in the first embodiment, the cassette **200** of the second embodiment includes the upper case **30**. Accordingly, the cassette **200** can help facilitate handling of the printing tape **101**, as in the first embodiment. As in the first embodiment, the rib **46**A in the second embodiment protrudes rearward from the opening edge **46**. Accordingly, the cassette **200** can help to suppress skewing of the ink ribbon **105**, as in the first embodiment.

Third Embodiment

[0180] Next, a cassette **300** according to a third embodiment of the present disclosure will be described with reference to FIGS. **20** and **21**. The cassette **300** includes the case **20**, as in the first embodiment described above. While the case **20** is configured of the upper case **30** and lower case **40**, the upper case **30** has been omitted from the drawing in FIG. **20**, as has been the lower cover **70** of the lower case **40**. In the cassette **300** of the third embodiment, parts and components having the same shape and function as those in the cassette **100** are designated with the same reference numerals to skip or simplify the description. In the cassette **300**, the maximum thickness of the main portion **84**A in the front-rear direction is greater than, less than, or the same as the distance in the front-rear direction between the maximum front surface area and the maximum rear surface area.

[0181] The cassette **300** is also for use in a printer **1**, and the manner of use is identical to that of the cassette **100** in the first embodiment. The cassette **300** differs from the cassette **100** in that the lower case **40** includes a center bridging portion **372**. The configurations in the cassette **300** other than the center bridging portion **372** are the same as the configurations of the cassette **100**.

[0182] The center bridging portion **372** is a wall that is part of the upper cover **80**. Specifically, the center bridging portion **372** is a portion of the upper cover **80** disposed in a space surrounded by the head front peripheral wall **84**, the head rear peripheral wall **85**, the head left peripheral wall **86**, and the head right peripheral wall **87**, i.e., the head opening **99**. The center bridging portion **372** extends in the up-down direction. The center bridging portion **372** extends leftward as extending rearward.

[0183] The center bridging portion **372** has a right-rear edge **372**A. The right-rear edge **372**A is connected to the head front peripheral wall **84** at the point P**1**. The right-rear edge **372**A extends from the point P**1** to the point P**2**. The right-rear edge **372**A is connected to either the head rear peripheral wall **85** or the head left peripheral wall **86** at the point P**2**. In the third embodiment, the right-rear edge **372**A is connected to the head left peripheral wall **86** at the point P**2**.

[0184] As in the second embodiment described above, the point P**1** in the third embodiment is located further leftward relative to the centerline C**1**. Further, the point P**2** is located further rearward relative to the front trisector V**1**. The point P**2** is at a position in the front-rear direction between the front edge and the rear edge of the head left peripheral wall **86**.

[0185] As described above, the lower case **40** of the third embodiment includes the center bridging portion **372**. With this configuration, if a force directed rearward acts on the lower edge of the head front peripheral wall **84**, the center bridging portion **372** receives the applied force. Therefore, this configuration suppresses the lower edge of the head front peripheral wall **84** from bending rearward relative to the upper edge of the head front peripheral wall **84**. Accordingly, the cassette **300** can help suppress skewing of the ink ribbon **105**.

[0186] As in the second embodiment described above, the point P**2** in the third embodiment is positioned further rearward relative to the front trisector V**1**. Accordingly, the cassette **300** can better help suppress skewing of the ink ribbon **105**, as in the second embodiment. As in the second embodiment, the point P**1** in the third embodiment is positioned further leftward relative to the centerline C**1**. Accordingly, the cassette **300** can ensure sufficient space in the head opening **99** while suppressing skewing of the ink ribbon **105**, as in the second embodiment.

[0187] As in the first embodiment described above, the tape insertion opening **44** of the third embodiment is formed in the intermediate wall **81**. Accordingly, the cassette **300** can help facilitate handling of the printing tape **101**, as in the first embodiment. As in the first embodiment, the cassette **300** of the third embodiment includes the upper case **30**. Accordingly, the cassette **300** can help facilitate handling of the printing tape **101**, as in the first embodiment. As in the first embodiment, the rib **46**A in the third embodiment protrudes rearward from the opening edge **46**. Accordingly, the cassette **300** can help suppress skewing of the ink ribbon **105**, as in the first embodiment.

[0188] The second embodiment and the third embodiment may be modified in various ways. For example, FIG. **22** illustrates a cassette **300**A according to a modification of the third embodiment. As illustrated in FIG. **22**, the point P**1** in the cassette **300**A is located at a position further rightward relative to the centerline C**1**. Similarly, the point P**1** may be located at a position further rightward relative to the centerline C**1** in the second embodiment.

[0189] FIG. **23** illustrates a cassette **300**B according to a modification of the second embodiment or the third embodiment. In the cassette **300**B illustrated in FIG. **23**, the lower case **40** includes both the lower bridging portion **272** and the center bridging portion **372**.

[0190] Other modifications of the second embodiment and the third embodiment will be described. In the second embodiment or the third embodiment, the point P**2** may be located at the same position as the front trisector V**1** in the front-rear direction, or may be located at a position further frontward relative to the front trisector V**1**. The second embodiment and the third embodiment may be modified in similar ways to the first embodiment as long as there are no inconsistencies.

[0191] While the invention has been described in conjunction with various example structures outlined above and illustrated in the figures, various alternatives, modifications, variations, improvements, and/or substantial equivalents, whether known or that may be presently unforeseen, may become apparent to those having at least ordinary skill in the art. Accordingly, the example embodiments of the disclosure, as set forth above, are intended to be illustrative of the invention, and not limiting the invention. Various changes may be made without departing from the spirit and scope of the disclosure. Therefore, the disclosure is intended to embrace all known or later developed alternatives, modifications, variations, improvements, and/or substantial equivalents.

<Remarks>

[0192] The cassettes **100**, **100**A, **100**B, **200**, **300**, **300**A, and **300**B are each an example of the “cassette”. The lower case **40** is an example of the “lower case”. The ribbon roll **107** is an example of the “ribbon roll”. The ink ribbon **105** is an example of the “ink ribbon”. The axis AX**1** is an example of the “first axis”. The take-up spool **109** is an example of the “take-up spool”. The front wall **141** is an example of the “front wall”. The rear wall **142** is an example of the “rear wall”. The left wall **143** is an example of the “left wall”. The right wall **144** is an example of the “right wall”. The head front peripheral wall **84** is an example of the “first inner wall”. The head rear peripheral wall **85** is an example of the “second inner wall”. The head left peripheral wall **86** is an example of the “third inner wall”. The head right peripheral wall **87** is an example of the “fourth inner wall”. The exposed area **92** is an example of the “exposed area”. The rib **97** is an example of the “protrusion”. The path-defining part **98**D is an example of the “first path-defining part”. The contact portion Z**3** is an example of the “first contact portion”. The path-defining part **98**G is an example of the “second path-defining part”. The contact portion Z**2** is an example of the “second contact portion”. The contact portion Z**1** is an example of the “third contact portion”. The virtual line S**1** is an example of the “first straight line”. The virtual line S**2** is an example of the “second straight line”. The lower wall **71** is an example of the “first wall”. The intermediate wall **81** is an example of the “second wall”. The outer peripheral wall **140** is an example of the “peripheral wall”. The tape insertion opening **44** is an example of the “insertion opening”. The printing tape **101** is an example of the “printing medium”. The upper case **30** is an example of the “upper case”. The tape roll **103** is an example of the “medium roll”. The axis AX**2** is an example of the “second axis”.

What is claimed is:

1. A cassette comprising:

a lower case having a box-like shape;

a ribbon roll accommodated in the lower case, the ribbon roll being a roll of an ink ribbon, the ribbon roll being rotatable about a first axis extending in an up-down direction; and

a take-up spool accommodated in the lower case, the take-up spool being configured to take up the ink ribbon from the ribbon roll,

wherein the lower case comprises:

a front wall extending in both the up-down direction and a left-right direction orthogonal to the up-down direction, the front wall having a right edge;

a rear wall extending in both the up-down direction and the left-right direction, the front wall and the rear wall being arranged in a front-rear direction orthogonal to both the up-down direction and the left-right direction;

a left wall extending in both the up-down direction and the front-rear direction;

a right wall extending in both the up-down direction and the front-rear direction, the left wall and the right wall being arranged in the left-right direction, the right wall having a front edge;

a first inner wall extending in both the up-down direction and the left-right direction, the first inner wall being positioned between the front wall and the rear wall, the first inner wall having a left edge and a right edge;

a second inner wall extending in both the up-down direction and the left-right direction, the second inner wall being positioned between the front wall and the rear wall and positioned further rearward relative to the first inner wall in the front-rear direction, the second inner wall having a left edge and a right edge;

a third inner wall extending in both the up-down direction and the front-rear direction, the third inner wall being positioned between the left wall and the right wall, the third inner wall having a front edge connected to the left edge of the first inner wall and a rear edge connected to the left edge of the second inner wall; and

a fourth inner wall extending in both the up-down direction and the front-rear direction, the fourth inner wall being positioned between the left wall and the right wall and positioned further rightward relative to the third inner wall in the left-right direction, the fourth inner wall having a front edge and a rear edge connected to the right edge of the second inner wall,

wherein the lower case has an exposed area in which the ink ribbon is exposed, the exposed area extending from a portion between the right edge of the front wall and the right edge of the first inner wall to a portion between the front edge of the fourth inner wall and the front edge of the right wall, and

wherein the lower case defines a conveying path along which the ink ribbon is conveyed in a conveying direction when the take-up spool takes up the ink ribbon from the ribbon roll, the conveying path including a first part, a second part, the exposed area, and a third part from an upstream side in the conveying direction, the first part passing between the left wall and the third inner wall, the second part passing between the front wall and the first inner wall, the third part passing between the fourth inner wall and the right wall,

the cassette further comprising:

a protrusion protruding rightward from a portion of the fourth inner wall other than the front edge and the rear edge of the fourth inner wall, the protrusion being configured to contact a widthwise center portion of the ink ribbon to cause the ink ribbon to be curved along a width of the ink ribbon so that the widthwise center portion of the ink ribbon protrudes rightward.

2. The cassette according to claim **1**,

wherein the lower case further comprises:

a first path-defining part provided on the rear edge of the fourth inner wall and defining the conveying path,

wherein the ink ribbon is configured to contact the first path-defining part in an entirety of the width of the ink ribbon to have a first contact portion with the first path-defining part, the first contact portion having a straight-line shape extending in the up-down direction when viewed in a longitudinal direction of the ink ribbon.

3. The cassette according to claim **2**,

wherein the lower case further comprises:

a second path-defining part provided further rearward relative to the first path-defining part, the second path-defining part being positioned on an opposite side of the ink ribbon from the first path-defining part when viewed in the up-down direction, the second path-defining part defining the conveying path,

wherein the ink ribbon is configured to contact the second path-defining part to have a second contact portion with the second path-defining part, the second contact portion being positioned downstream of the first contact portion in the conveying direction,

wherein the ink ribbon is configured to contact the protrusion to have a third contact portion with the protrusion, and

wherein the second contact portion and the third contact portion define a first straight line passing through the second contact portion and the third contact portion, the first straight line passing through a point positioned to a left of the first contact portion when viewed in the up-down direction.

4. The cassette according to claim **3**,

wherein the conveying path further includes a fourth part positioned downstream of the third part in the conveying direction, the fourth part passing between the right wall and a center of the take-up spool, and

wherein the first contact portion and the take-up spool define a second straight line passing through the first contact portion and tangential to an outer circumferential surface of the take-up spool at a position rightward of the center of the take-up spool, the second straight line passing through a point positioned to a right of the second contact portion when viewed in the up-down direction.

5. The cassette according to claim **1**,

wherein the lower case further comprises:

a first wall extending in both the front-rear direction and the left-right direction;

a second wall extending in both the front-rear direction and the left-right direction, the first wall and the second wall being arranged in the up-down direction; and

a peripheral wall extending between the first wall and the second wall, the peripheral wall comprising the front wall, the rear wall, the left wall, and the right wall,

wherein the second wall has an insertion opening allowing a printing medium to be inserted into the lower case therethrough,

wherein the lower case further defines a conveying path along which the printing medium is conveyed in a conveying direction, the conveying path of the printing medium including the insertion opening, a first part, a second part, and the exposed area from an upstream side in the conveying direction of the printing medium, the first part of the conveying path of the printing medium passing between the left wall and the third inner wall, the second part of the conveying path of the printing medium passing between the front wall and the first inner wall, and

wherein the ink ribbon and the printing medium are configured to be superimposed with each other in the exposed area.

6. The cassette according to claim **5**, further comprising:

an upper case having a box-like shape, the upper case being positioned further upward relative to the lower case, wherein the first wall is positioned further downward relative to the second wall; and

a medium roll accommodated in the upper case, the medium roll being a roll of the printing medium, the medium roll being rotatable about a second axis extending in the up-down direction, wherein the printing medium is configured to be drawn off the medium roll and to be inserted into the lower case through the insertion opening.

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